

Orchard Pollination Planting Grid and West Windbreak

Prairie Plants — inventory of October 4, 2026 — orchard plot 237 × 600 ft plus a 35 ft west windbreak — 13 positions per row — 18 ft in-row (E–W) × 20 ft rows (N–S), triangular offset

Sources: [Inventory_PrairiePlants.xlsx](#) (what is actually at Prairie, saved October 4, 2026), matched to [Trees.xlsx](#), Fruit sheet (cultivar, rootstock, fertility, and flowering-group data, including the September 24 bloom research). Only trees found at Prairie are planned. This standalone page is a snapshot, not a live spreadsheet connection.

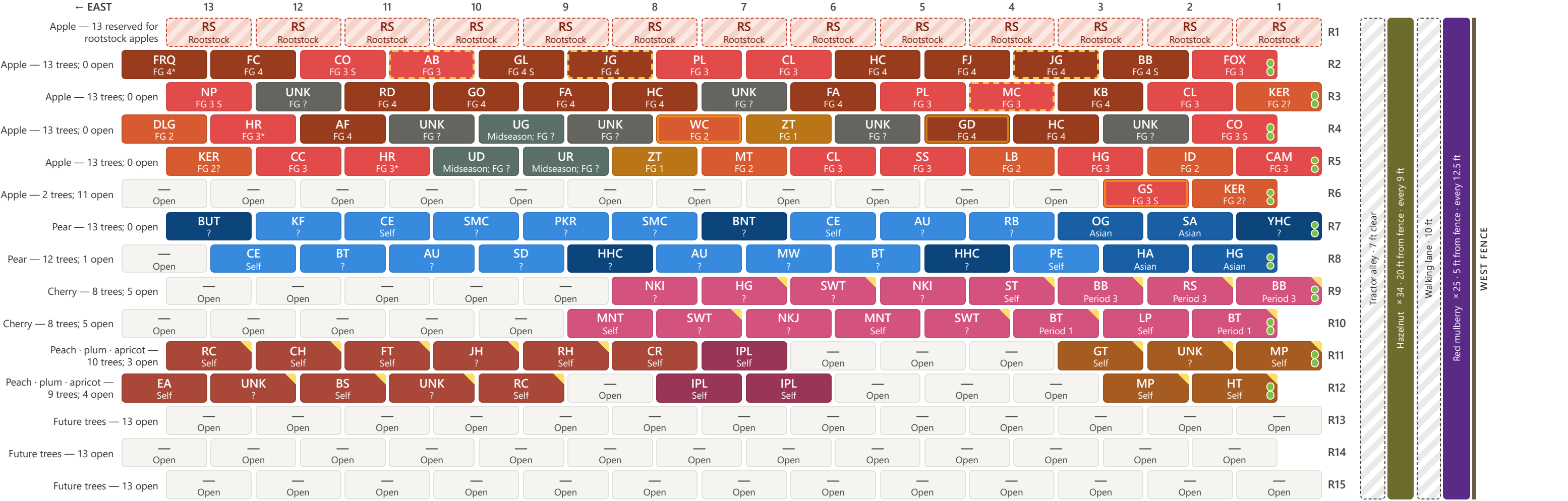
ORCHARD PLOT 237 × 600 ft	WEST WINDBREAK 35 ft, 2 rows	TRACTOR ALLEY 7 ft clear	TREE SPACING 18 ft in-row	ROW SPACING 20 ft N–S	DIAGONAL NEIGHBOR 21.9 ft	ORCHARD TREES 114	OPEN POSITIONS 68	ROWS 15 (S→N)	WINDBREAK · HEDGE · GUILDS 59 + 19 + 40
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Width, west to east: the 35 ft windbreak strip (2 plant rows with a 10 ft walking lane between them, then the tractor alley), then the 237 ft orchard plot. C1 trunks stand 3 ft inside the orchard's west line, 38 ft from the fence. 13 staggered positions span 225 ft, leaving 9 ft to the east line beside the offset rows' C13 trees (18 ft beside the others); a 14th position would need 243 ft. East of the orchard: 28 ft of open ground for turning at the row ends, the 22 ft driveway, then 10 ft to the east fence (332 ft from the west fence), with the elderberry hedge 5 ft inside it. Depth: 15 rows span 280 ft between the outer row centers. With the windbreak ends, the planted block uses about 300 ft of the 600 ft depth, leaving about 300 ft for headlands and future rows. Distances are tree centers, before mature-canopy clearance; the 18 × 20 ft spacing is a planning assumption, not a mature-spacing recommendation for every rootstock.

Planting Map — South at Top, West Fence on the Right

South is up, the west fence and windbreak are on the right, and east is on the left: the view from the north end looking south. Rows start at the west alley (C1, on the right) and fill toward the east (left), so open spots are at the east ends, ready to fill in from the east later, except that in R11–R12 the apricots are right-justified, plums are centered, and peaches are left-justified, so the open spots there are between the blocks. Rows from south to north (top to bottom): apples in 5 rows (13 + 13 + 13 + 13 + 2) plus reserved R1 (13 spots); pears in 2 rows (13 + 12); cherries in 2 rows (8 + 8); apricots at the west end (3 + 2), plums in the center (1 + 2), and peaches at the east end (6 + 5) share R11–R12; R13–R15 are open for trees you buy or graft later. R1 is held for 13 grafted rootstock apples. The C1 trees in R2–R12 have berry guilds on their alley side (green dots); exact spots are on the site plan. The windbreak band is schematic; exact positions are in the windbreak table and site plan, which also shows the elderberry hedge along the east fence. Full names, tags, and notes are listed by position in the inventory.

▲ SOUTH — the closest juniper or red cedar is 200+ ft beyond this end



▼ NORTH

- Apple FG 1 Apple FG 2 Apple FG 3 Apple FG 4 Apple — relative bloom only; FG unverified Apple — unidentified or bloom unconfirmed Pear — European Pear — Asian Pear — Perry Cherry Apricot Peach Plum Reserved: grafted rootstock apple (RS)
- Windbreak: Red mulberry Windbreak: Hazelnut Berry guild on the alley side of the C1 tree

▲ Dashed border = **triploid**, excluded as a pollen donor and placed near two different diploid cultivars. ★ Gold glow = **good pollen donor in the workbook**, not a guarantee of compatibility with every nearby tree.

▼ Yellow corner = **Pseudomonas (bacterial canker) positive, untreated** in the inventory. When one tag covers several trees and only some tested positive, the inventory does not say which, so all of them are flagged.

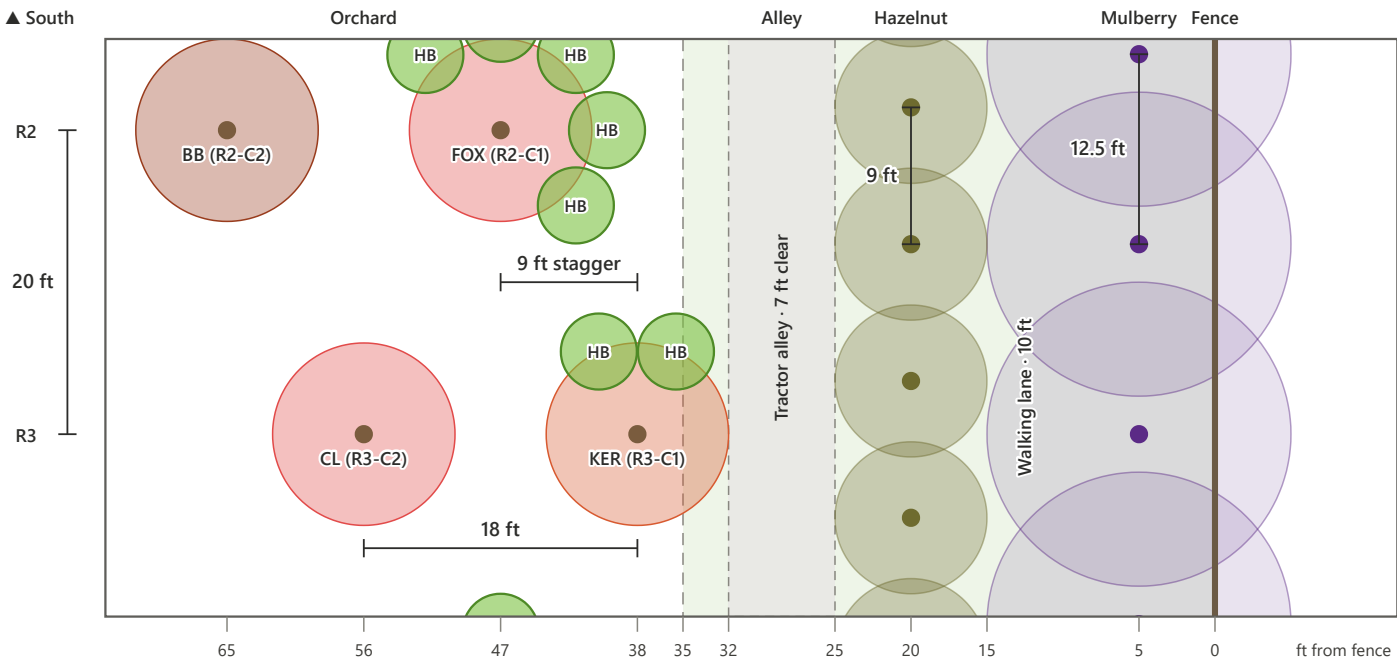
S = self-fertile according to the planning notes; **?** = information missing or needing confirmation; **UNK** / **SWT** = unidentified tree (not counted as a pollen donor until identified). Different trees of the same cultivar, including color sports, are not counted as different pollen partners.

Apple **FG** means flowering group, not harvest season. * after a group number means sources differ; the planning value follows Orange Pippin. **FG 2?** marks a provisional estimate. **Midseason; FG ?** records nursery bloom guidance without inventing a numbered group.

West Windbreak (35 ft Strip) and East Elderberry Hedge

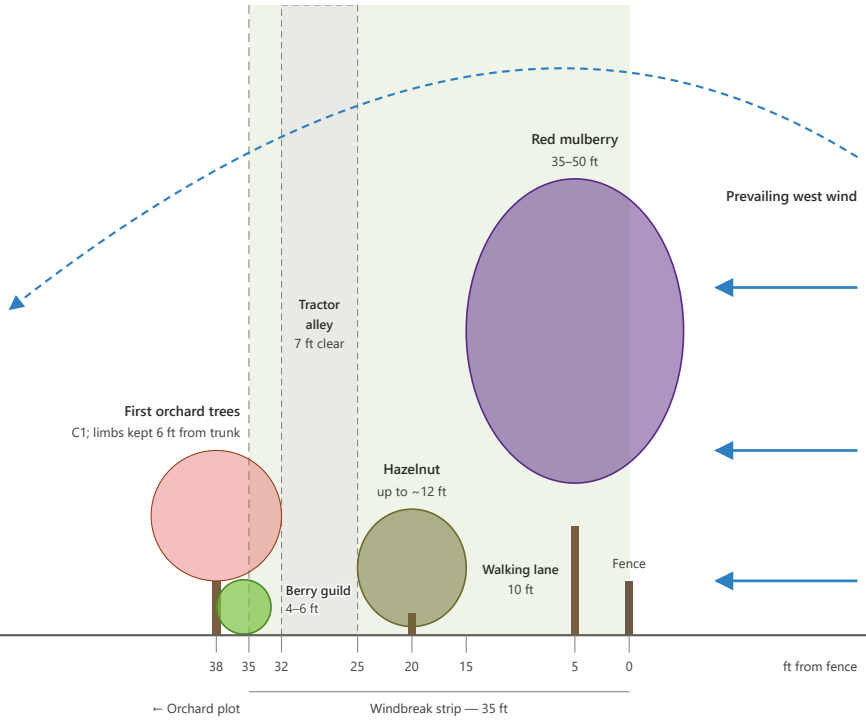
Two north–south rows step down from the west fence toward the orchard: red mulberry (tallest) along the fence and hybrid hazelnut 15 ft east of it, so the prevailing west wind is pushed up and over the fruit trees. A 10 ft walking lane under the mulberries leads to the hazelnuts for picking. The strip is added west of the 237 ft orchard plot; the orchard was not reduced. Both rows spread the plants you have along the orchard and reach at least 6 ft past R1 and R15. The 41 berry bushes came out of the strip to make room for the lane and now form guilds on the alley side of the C1 orchard trees in R2–R12. The elderberry hedge runs 5 ft inside the east fence beside the driveway at its proper spacing from the south end, with 19 positions marked for plants to add later. East–west positions are measured from the west fence and north–south positions from the R1 row line. General notes are beside the site plan.

Blow-up plan view, to scale: a 38 ft stretch of the windbreak beside the first apple rows (south up, west on the right)



Stretch shown: 14–52 ft north of the R1 line. Green circles: berry guild bushes around the C1 trees (HB honeyberry).

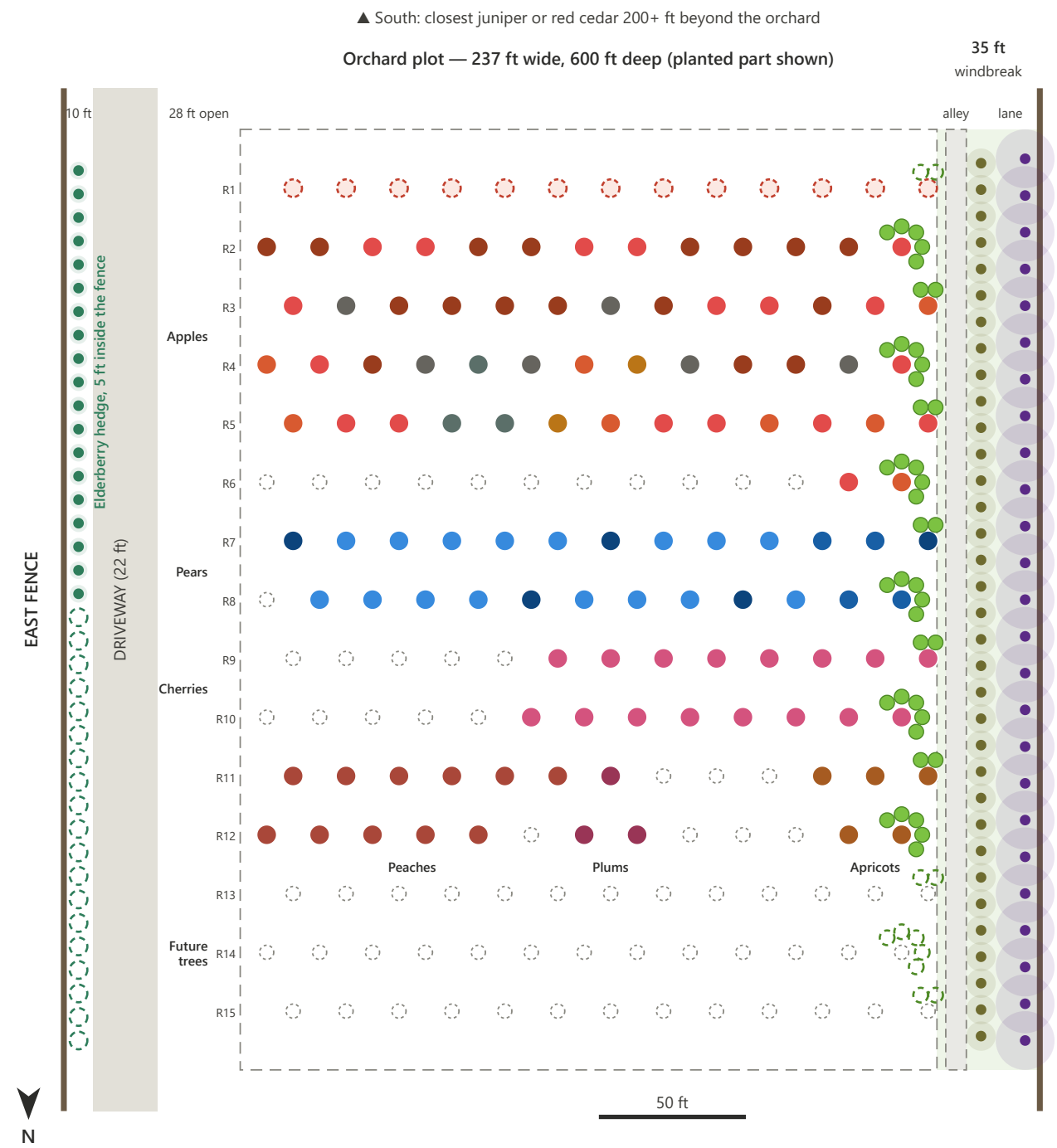
Profile, looking south: plant heights step down from the fence toward the orchard



Layer (west → east)	Plant	Count	From west fence	In-row spacing	First plant	Last plant	Mature size	Notes
1. Tall layer on the west fence	Red mulberry Morus rubra	25	5 ft	12.5 ft (12 ft 6 in)	10 ft south of the R1 line	10 ft north of the R15 line	35–50 ft tall; 35–40 ft wide naturally; keep about 20 ft	Main wind barrier. 12.5 ft is within the 6–15 ft in-row range UNL gives for deciduous windbreak trees. It is a tree, usually 15–70 ft tall (USDA Forest Service), but grown in the open it tends to have a short trunk with large, low, spreading limbs (Purdue) and it sprouts from the roots (USDA Forest Service), so left alone it can look like a wide, many-stemmed bush. Seedlings are usually male or female, so expect some fruitless trees (MOBOT). Red and white mulberry cross and are hard to tell apart (Purdue); white (Russian) mulberry, long planted in western windbreaks, is often a smaller, many-branched tree. Prune the east side back to about 10 ft from the trunk so it stays over the walking lane, not the hazelnuts. Ripe mulberries may also draw birds away from the cherries.
2. Shrub layer beside the alley	Hazelnut Corylus hybrids	34	20 ft	9 ft	8.5 ft south of the R1 line	8.5 ft north of the R15 line	Up to about 12 ft tall; nearly as wide as tall; keep about 10 ft	15 ft from the mulberries; USDA's hybrid hazelnut guide keeps taller tree rows at least 15 ft away so they do not shade out the hazels. Hybrids grow as multi-stemmed bushes, not trees, up to about 12 ft tall and nearly as wide, which can take ten years or more (UMHDI). 9 ft apart is within the 6–9 ft UMHDI gives for nut production; layer low stems into the gaps to close the row sooner. Pick by hand as soon as the nuts loosen in their husks (UMHDI); seedlings ripen over a month or more, late August into October (UMHDI harvest bulletin), so plan several passes. After about 12 years, cut a few bushes to the ground each winter to keep them small enough to pick (UMHDI).
East hedge beside the driveway	American elderberry Sambucus canadensis	19 planted + 19 to add	327 ft (5 ft inside the east fence)	8 ft	6 ft south of the R1 line	138 ft north of the R1 line (last planted); open positions to 10 ft north of the R15 line	6–12 ft tall; 6–8 ft or wider; keep about 6 ft	A separate, tighter hedge in full sun between the driveway and the east fence. WVU suggests at least 8–9 ft apart in the row for fruit production (plants with a 6–8 ft spread). Several Berry-sheet elderberries need another variety, so mix the named cultivars along the row. Fill the open positions with dormant hardwood cuttings, which WVU says root easily. Prune to about 6 ft wide so it stays 2 ft off the driveway, mow suckers, and keep plowed snow off young plants. At 6–12 ft tall it shelters roughly 30–60 ft to the west (about 5 times its height, UNL): the driveway, the headland, and at full height the orchard's east edge, 55 ft away.

Site Plan — To Scale, South Up

To scale, south up and west on the right, matching the planting map (the view from the north end looking south). East–west distances are from the west fence; north–south distances are from the R1 row line. Large dots are orchard trees in fruit-type colors, dashed red rings are positions reserved for grafted rootstock apples, and dashed gray circles are open positions. Small dots are windbreak plants (purple red mulberry, olive hazelnut) with their managed mature width shaded; the 10 ft walking lane runs between the two rows. Lime dots beside the C1 trees are the berry guilds, and dashed lime rings are guild positions waiting for the R1 and future-row trees. The green dots between the driveway and the east fence are the elderberry hedge, and its dashed rings mark positions for plants to add later. East of the orchard are 28 ft of open ground for turning at the row ends, the driveway, the hedge, and the east fence. The plot is 600 ft deep; only the planted 300 ft is drawn. The block can slide north or south, but keeping it toward the north end adds distance from the junipers. Hover over a dot for its position.



Windbreak, hedge, and berry guild notes

Row spacing. The two rows are 15 ft apart: USDA's hybrid hazelnut guide keeps taller tree rows at least 15 ft from the hazels, and UNL's windbreak guide typically spaces rows 12–16 ft apart. The berry bushes moved out of the strip to make this room and now form guilds around the C1 trees.

Walking lane. The 10 ft from the mulberry trunks to the west edge of the hazelnuts (5–15 ft from the fence) is for walking and picking. Train each mulberry to one trunk, cut the root sprouts every year, and raise the crown a little each winter so you can walk under it. Remove low limbs while they are small: mulberry cuts on limbs 2 in or thicker may not heal (Ask Extension), and pruning in late fall or winter avoids bleeding (MOBOT). Mulberries are picked by shaking branches over a tarp (Ask Extension), so keep the lane mowed or mulched and clear. Pick the hazelnuts' west side from the lane and their east side from the tractor alley.

Tractor alley. The hazelnut row's east edge is 25 ft from the fence and the first orchard trunks are 38 ft from it, leaving 13 ft at ground level. Keep the alley-side limbs of the row-end orchard trees within 6 ft of the trunk to hold a 7 ft clear lane. If the hazelnuts reach their full 12 ft width, the alley narrows to 6 ft (your minimum is 6 ft). C1 trees in the even rows stand another 9 ft east.

Protection. UNL reports that a moderately dense windbreak cuts a 20 mph wind to about 5 mph out to 5 times its height. Mature mulberries (35–50 ft) would reach roughly 175–250 ft into the 237 ft orchard; in the first years the hazelnuts mainly shelter the west end. Leafless rows are thinner in early spring, when the stone fruit bloom.

Length. Wind wraps around the ends, so UNL recommends extending a windbreak 100–200 ft beyond the area it protects. The mulberry and hazelnut rows cover R1–R15 plus at least 6 ft at each end. The east hedge starts at the south end; 19 more elderberries at the same spacing finish it to the north end. The plot has room to extend all the rows north and south later.

Junipers. Junipers and Eastern red cedar host cedar–apple rust, and UMN advises keeping them a few hundred yards from apples; spores can travel more than a mile. Yours are south of the orchard, the closest 200+ ft away, and the apples are at the south end as requested: R1 is about 200+ ft from the closest juniper, and the pears (occasional hosts of a related rust) about 320+ ft. UMN suggests inspecting nearby junipers and red cedars in late winter or early spring and pruning out brown, woody galls before the orange, gelatinous structures form, and notes that the rust is only a minor problem on resistant or partially resistant apples. Do not add junipers or red cedars to this windbreak, and watch the apples for orange leaf spots in late spring.

Pollination. Hazelnuts are wind-pollinated and cannot pollinate themselves (UMHDI); the seedling hybrids should pollinate one another. Several named elderberries need another variety (Berry sheet), so mix the cultivars along the hedge.

Berry guilds. Each C1 tree in R2–R12 has berry bushes on its alley side: an arc of 5 around the south and west of each even-row tree, which stands 9 ft farther from the alley, and 2 on the south side of each odd-row tree. They sit 6–7 ft from the trunks and about 5 ft apart, all east of the alley (the nearest bush edge is 33 ft from the fence), with a walking gap of at least 4.5 ft between neighboring guilds. South to north: honeyberry ×11 (R2–R4), red currant ×7 (R4–R6), gooseberry ×3 (R6–R7), Viking aronia ×2 (R8), jostaberry ×17 (R8–R12). Left over: 1 jostaberry. The 11 dashed positions around the C1 spots in R1, R13, R14, and R15 wait for those trees. The south side of each tree gets the most sun. Honeyberries tolerate partial shade and need at least two varieties that pollinate each other (UAF Extension), so mix their varieties; currants and gooseberries tolerate partial shade but fruit more in full sun (UMN). Comfrey could fill the east side of each guild, in the tree row, but its deep roots make it difficult to eradicate once established (WSU). Keep one side of each tree open for a ladder.

Blister rust. White pine blister rust attacks five-needle (white) pines and uses currants and gooseberries as alternate hosts (WSU). Your white pines died, and ponderosa and Austrian pines are not five-needle pines, so the guild currants and gooseberries put no pines at risk; UMN also lists jostaberry as resistant. Avoid planting white pines near them later.

October 4, 2026 Update — What Changed

- Planned from what is at Prairie: 114 fruit trees from Inventory_PrairiePlants.xlsx, each matched to its Trees.xlsx purchase record. 12 unidentified trees are placed with their fruit type but not counted as pollen donors. The September plan used the full purchase list (122 fruit trees and 32 nut trees).
- Orchard plot set to 237 × 600 ft: the measured 232 ft plus 5 ft from the open ground to the east, so the last trees have 9 ft (half the tree spacing) to the east line. About 28 ft remains between the orchard and the driveway for turning. The September plan used 225 ft plus a 5 ft buffer.
- A 35 ft windbreak strip was added between the orchard and the west fence without shrinking the orchard: rows of red mulberry and hazelnut step down from west to east, followed by a 7 ft clear tractor alley.
- The elderberries moved out of the west windbreak to their own hedge 5 ft inside the east fence, beside the driveway: 8 ft apart for a tighter grouping and some shelter from east winds. The 19 plants fill the south 144 ft; 19 more positions are marked to finish the hedge.
- To leave room to walk and pick, the west strip is now two rows: red mulberry 5 ft and hazelnut 20 ft from the fence, 15 ft apart as USDA's hybrid hazelnut guide advises, with a 10 ft walking lane under the mulberries.
- The berry bushes moved out of the west strip into guilds on the alley side of the C1 trees in R2–R12, following your sketch: 40 bushes (honeyberry ×11, red currant ×7, gooseberry ×3, Viking aronia ×2, jostaberry ×17), with 1 left over. 11 more positions wait for the trees in R1 and the future rows.
- The two white pines died (owner report), so white pine blister rust, which alternates between five-needle pines and currants or gooseberries, is no longer a concern for the currants and gooseberries; ponderosa and Austrian pines are not five-needle pines.
- The main orchard holds fruit trees only. Nut trees, rootstocks, and the two Boise trees are listed separately, and the hazelnuts moved to the windbreak.
- Rows start at the west alley and fill eastward, so open positions are at the east ends, except that in R11–R12 the apricots are right-justified, plums are centered, and peaches are left-justified, so the open spots there are between the blocks.
- Rows by fruit type: apples in 5 rows (13 + 13 + 13 + 13 + 2) plus reserved R1 (13 spots); pears in 2 rows (13 + 12); cherries in 2 rows (8 + 8); apricots at the west end (3 + 2), plums in the center (1 + 2), and peaches at the east end (6 + 5) share R11–R12; R13–R15 are open for trees you buy or graft later. The cherries use the two rows you chose, and the shared rows R11–R12 follow your layout; the orchard keeps all 15 rows.
- Added R1, a full apple row at the top of the map with 13 spots reserved for grafted rootstock apples. The inventory has 15 rootstocks (2 EMLA 7, 12 M-111, 1 unknown).
- The fruit types run south to north as apples, pears, cherries, then apricots, plums, and peaches sharing R11–R12, with the apples at the top of the map; the September plan had apples, pears, cherries, apricots, peaches, and plums, each in its own rows. The apples are about 200+ ft from the closest juniper, closer than UMN advises; the windbreak notes list ways to limit cedar–apple rust.
- J.H. Hale now relies on other non-Elberta peaches nearby because Veteran, its earlier partner, was not found at Prairie.
- The map is drawn with south at the top and the west fence on the right, the view from the north end looking south; the site plan and windbreak cross-section match. The earlier grid put south at the top but kept west on the left, which mirrored the real layout.

Prairie Inventory vs. Purchase List

At Prairie: 114 fruit trees (12 unidentified), 15 apple rootstocks, 20 nut trees, 59 west windbreak plants, 19 elderberries for the east hedge, and 41 berry bushes for the C1 guilds. Compared with the Trees.xlsx Fruit sheet, 22 purchased fruit trees were not found at Prairie, 4 more than purchased were found for a few varieties, and 2 trees remain in Boise. Some missing trees may be among the unidentified ones. The table lists only the differences; the line-by-line match is below it.

Purchased vs. found — differences only

Purchase record(s), Trees.xlsx Fruit sheet	Purchased	At Prairie	Elsewhere	Difference	Notes
Aztec Fuji Apple; Fruit rows 4, 5	3	2		-1	2 of 3 purchased found (row 4: 1 semi-dwarf; row 5: 2 dwarf); the inventory does not say which. May be among the 6 unidentified apples at Prairie.
Braeburn Apple; Fruit rows 7, 8	2	1		-1	1 of 2 purchased Braeburns found. The tag says Dwarf, so it is matched to the Lowes tree with no recorded rootstock (row 8) rather than the M-111 tree (row 7). May be among the 6 unidentified apples at Prairie.
Cortland Apple; Fruit row 10	5	3		-2	3 of 5 purchased found. May be among the 6 unidentified apples at Prairie.
Dolgo Crabapple Apple; Fruit row 13	2	1		-1	1 of 2 purchased found. May be among the 6 unidentified apples at Prairie.
Empire Apple; Fruit row 14	1	0		-1	Not in the Prairie inventory. May be among the 6 unidentified apples at Prairie.
Gala Apple; Fruit rows 19, 20	2	1		-1	1 of 2 purchased found. May be among the 6 unidentified apples at Prairie.
Golden Russet Apple; Fruit row 22	1	0		-1	Not in the Prairie inventory. May be among the 6 unidentified apples at Prairie.
Honeycrisp Apple; Fruit row 28	3	2		-1	2 of 3 purchased found. May be among the 6 unidentified apples at Prairie.
Kerr Crab Seedling Apple; Fruit row 33	2	3		+1	3 found, but 2 were purchased. All are seed-grown, so each may bloom differently.
Sweet Sixteen Apple; Fruit row 41	2	1		-1	1 of 2 purchased found. May be among the 6 unidentified apples at Prairie.
Zestar Apple; Fruit row 45	3	2		-1	2 of 3 purchased found. May be among the 6 unidentified apples at Prairie.
Chinese (Mormon) Apricot; Fruit row 46	1	0	1	0	Still in Boise (girdled, mulched); not planned for Prairie.

Purchase record(s), Trees.xlsx Fruit sheet	Purchased	At Prairie	Elsewhere	Difference	Notes
Moorpark Apricot; Fruit row 49	1	2		+1	2 found, but 1 Moorpark was purchased. Patterson (row 50) was not found; one of these may be Patterson.
Patterson Apricot; Fruit row 50	1	0		-1	Not in the Prairie inventory. May be the unidentified apricot at Prairie.
Black Tartarian Cherry; Fruit rows 55, 56	4	2		-2	2 of 4 purchased found. May be among the 3 unidentified sweet cherries at Prairie.
Craig's Crimson Cherry; Fruit row 57	1	0	1	0	Still in Boise (girdled, mulched); not planned for Prairie.
Regina Cherry; Fruit row 62	1	0		-1	Not in the Prairie inventory. May be among the 3 unidentified sweet cherries at Prairie.
Sam Sweet Cherry; Fruit row 64	1	0		-1	Not in the Prairie inventory. May be among the 3 unidentified sweet cherries at Prairie.
Hybrid Hazelnut / Jefferson Filbert Hazelnut / Theta Filbert Hazelnut / Hybrid Hazelnut (controlled-cross) Nut; Fruit rows 70, 71, 76, 118	77	34		-43	Listed as Jefferson Filbert, but the purchase list has 75 Forestag hybrid seedlings, 1 Jefferson, and 1 Theta. Most are probably hybrids; if the Jefferson and Theta can be identified, plant them side by side (Theta pollinizes Jefferson).
Pecan (Native Hardy) Nut; Fruit row 74	10	4		-6	4 of 10 purchased found.
Shellbark Hickory Nut; Fruit row 75	10	0		-10	Not in the Prairie inventory.
Contender Peach; Fruit row 79	2	1		-1	1 of 2 purchased found. May be among the 2 unidentified peaches at Prairie.
Elberta Peach; Fruit row 80	2	1		-1	1 of 2 purchased found. May be among the 2 unidentified peaches at Prairie.
Reliance Peach; Fruit rows 85, 86, 87	3	2		-1	2 of 3 purchased found. May be among the 2 unidentified peaches at Prairie.
Veteran Peach; Fruit row 88	1	0		-1	Not in the Prairie inventory. May be among the 2 unidentified peaches at Prairie.
Anjou Pear; Fruit row 89	2	3		+1	Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased.
Hendre Huffcapp European Perry Pear Pear; Fruit row 95	1	2		+1	2 found, but 1 was purchased. Thorn (row 107) was not found; one of these may be Thorn.
Thorn European Perry Pear Pear; Fruit row 107	1	0		-1	Not in the Prairie inventory.
Italian Plum / Italian Seedling Plum; Fruit rows 109, 110	4	3		-1	3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings.
Sambucus canadensis Elderberry; Fruit row 115	25	19		-6	Planted as a hedge along the east fence by the driveway. The purchase list also has 9 named elderberries on the Berry sheet (American, Bob Gordon ×2, Marge ×2, Perl, Ranch, Wyldewood, York). The 19 at Prairie are not identified by cultivar.
Amelanchier canadensis Serviceberry; Fruit row 117	25	0		-25	Not in the Prairie inventory.

Show every inventory line used in this plan and where it goes

Inventory row	Prairie inventory	Qty	Matched purchase record(s)	Plan / positions	Notes
3	Currant — Red Berry Tag Currant	7	Berry rows 4–7: 5 currants (1 red, 3 black, 1 white)	C1 berry guilds, R4–R6	7 currants listed as red at Prairie vs. 5 purchased (1 red, 3 black, 1 white). Self-fertile per the Berry sheet.
4	Elderberry	19	Row 115: Sambucus canadensis (25)	East hedge, 5 ft inside the east fence	Planted as a hedge along the east fence by the driveway. The purchase list also has 9 named elderberries on the Berry sheet (American, Bob Gordon ×2, Marge ×2, Perl, Ranch, Wyldewood, York). The 19 at Prairie are not identified by cultivar.
5	Gooseberry Tag GooseBerry	3	Berry rows 16–20: 5 gooseberries (Captivator, Hinnonmaki Red ×2, Invicta, Pixwell)	C1 berry guilds, R6–R7	3 at Prairie, 2 fewer than purchased. Self-fertile per the Berry sheet.
7	Honeyberry Tag Hny Bry	11	Berry rows 26–31: 12 honeyberries (Blue Diamond ×3, Blue Palm ×3, Indigo Treat ×2, Cinderella ×2, Maxine, Sunrise)	C1 berry guilds, R2–R4	11 at Prairie, 1 fewer than purchased. The Berry sheet says they need another honeyberry, so plant them as one group and mix the varieties.

Inventory row	Prairie inventory	Qty	Matched purchase record(s)	Plan / positions	Notes
8	Josta — Giant Tag JstB-G	4	Berry rows 32–34: 7 jostaberries (6 black, 1 red)	C1 berry guilds, R8–R12 (1 left over)	18 jostaberries at Prairie on lines 8–10, 11 more than purchased. Self-fertile per the Berry sheet.
9	Josta — Red Tag JstB-R	10	Berry rows 32–34: 7 jostaberries (6 black, 1 red)	C1 berry guilds, R8–R12 (1 left over)	18 jostaberries at Prairie on lines 8–10, 11 more than purchased. Self-fertile per the Berry sheet.
10	Josta — Regular Tag JstB	4	Berry rows 32–34: 7 jostaberries (6 black, 1 red)	C1 berry guilds, R8–R12 (1 left over)	18 jostaberries at Prairie on lines 8–10, 11 more than purchased. Self-fertile per the Berry sheet.
11	Mullberry — Red	25	Row 116: Morus rubra (25)	West windbreak, red mulberry row	
13	Viking Tag Viking	2	Berry rows 2–3: 2 Viking aronia (chokeberry)	C1 berry guilds, R8	Matches the purchase list. Self-fertile per the Berry sheet.
15	Hazlenut — Jefferson Filbert	34	Row 70: Hybrid Hazelnut (25); Row 71: Jefferson Filbert Hazelnut (1); Row 76: Theta Filbert Hazelnut (1); Row 118: Hybrid Hazelnut (controlled-cross) (50)	West windbreak, hazelnut row	Listed as Jefferson Filbert, but the purchase list has 75 Forestag hybrid seedlings, 1 Jefferson, and 1 Theta. Most are probably hybrids; if the Jefferson and Theta can be identified, plant them side by side (Theta pollinizes Jefferson).
26	Apple — Arkansas Black Tag AL-Ark Blk	1	Row 2: Arkansas Black Spur (FT015) (1)	R2-C10	
27	Apple — Braeburn Dwarf Tag AL-Brae Dwf	1	Row 8: Braeburn (1); Row 7: Braeburn (1)	R2-C2	1 of 2 purchased Braeburns found. The tag says Dwarf, so it is matched to the Lowes tree with no recorded rootstock (row 8) rather than the M-111 tree (row 7).
28	Apple — Campfield Cide Tag AL-Campfield	1	Row 9: Campfield Cider Apple (1)	R5-C1	
29	Apple — Cortland Tag AL-Cort/ AL-Cortland	3	Row 10: Cortland (5)	R2-C6 , R3-C2 , R5-C6	3 of 5 purchased found.
30	Apple — Cox Orange Pippin Tag AL-O Pip	2	Row 11: Cox's Orange Pippin (2)	R2-C11 , R4-C1	
31	Apple — Crabapple, Dolgo Tag AL-Dolgo	1	Row 13: Dolgo Crabapple (2)	R4-C13	1 of 2 purchased found.
32	Apple — Crabapple, Kerr Tag AL-Kerr Crab	3	Row 33: Kerr Crab Seedling (2)	R3-C1 , R5-C13 , R6-C1	3 found, but 2 were purchased. All are seed-grown, so each may bloom differently.
33	Apple — Crabapple, Wickson Tag AL-Wickson cr	1	Row 44: Wickson Crab (1)	R4-C7	
34	Apple — Crimson Crisp Tag AL-Crimson	1	Row 12: CrimsonCrisp (1)	R5-C12	
35	Apple — Crispin Tag AL-Crisp	1	Row 37: Mutsu Crispin (1)	R3-C4	
36	Apple — Delicious, Golden Tag AL-Gold D	1	Row 21: Golden Delicious (1)	R4-C4	
37	Apple — Delicious, Red Tag AL-Red Del	1	Row 40: Red Delicious (1)	R3-C11	
38	Apple — EMLA7 Apple Rootstock Tag AL-Rtsk-EMLA7	2	No purchase record matched	Graft first; reserved row R1 holds 13 of them	Ungrafted EMLA 7 rootstock, for grafting.
39	Apple — Foxwhelp Cider Tag AL-Foxwhelp	1	Row 15: Foxwhelp Cider Apple (1)	R2-C1	
40	Apple — Franklin Cider Tag AL-Frank C	1	Row 16: Franklin Cider (1)	R2-C12	
41	Apple — Frequin Rouge Tag AL-Frequin	1	Row 17: Frequin Rouge Cider Apple (1)	R2-C13	
42	Apple — Fuji Tag AL-Fuji	1	Row 18: Fuji (1)	R2-C4	Same tag text (AL-Fuji) as the Early Fuji; check the variety label when planting.
43	Apple — Fuji Aztec Tag AL-Az Fuji	2	Row 4: Aztec Fuji (1); Row 5: Aztec Fuji (2)	R3-C6 , R3-C9	2 of 3 purchased found (row 4: 1 semi-dwarf; row 5: 2 dwarf); the inventory does not say which.
44	Apple — Fuji, Early Tag AL-Fuji	1	Row 3: Auvil Early Fuji (1)	R4-C11	Same tag text (AL-Fuji) as the Fuji; check the variety label when planting.
45	Apple — Gala Tag AL-Gala	1	Row 19: Gala (1); Row 20: Gala (1)	R2-C9	1 of 2 purchased found.

Inventory row	Prairie inventory	Qty	Matched purchase record(s)	Plan / positions	Notes
46	Apple — Goldrush Tag AL-Gld Rsh	1	Row 24: GoldRush (1)	R3-C10	
47	Apple — Granny Smith Tag AL-Gran S	1	Row 25: Granny Smith (1)	R6-C2	
48	Apple — Harrison Tag AL-Harrison	2	Row 26: Harrison (1); Row 27: Harrison Cider Apple (1)	R4-C12 , R5-C11	The workbook lists one Harrison (row 26, M-111) and one Harrison Cider (row 27, semi-dwarf), the same cultivar. The tags do not say which is which, so both use the row 26 cultivar data.
49	Apple — Honeycrisp Tag AL-Hon Cris	2	Row 28: Honeycrisp (3)	R2-C5 , R3-C8	2 of 3 purchased found.
50	Apple — Honeycrisp, Super Red Tag AL-Hny Crisp	1	Row 29: Honeycrisp (Super Red) (1)	R4-C3	
51	Apple — Hudsons Golden Gem Tag AL-Gold Gem	1	Row 30: Hudson's Golden Gem (1)	R5-C3	
52	Apple — Idared Tag AL-Idared	1	Row 31: Idared (1)	R5-C2	
53	Apple — Jonagold Tag AL-Jon G	2	Row 32: Jonagold (2)	R2-C3 , R2-C8	
54	Apple — Kingston Black Tag AL-G Orch-7	1	Row 34: Kingston Black (1)	R3-C3	Tag AL-G Orch-7 (Grandpa's Orchard, M-7) matches the workbook.
55	Apple — Liberty Tag AL-Liberty	1	Row 35: Liberty (1)	R5-C4	
56	Apple — M 111 Rootstock Tag AL-Rstk M111	12	No purchase record matched	Graft first; reserved row R1 holds 13 of them	Ungrafted M-111 rootstock, for grafting.
57	Apple — McIntosh Tag AL-Mc In	1	Row 36: Macintosh (1)	R5-C7	
58	Apple — Newton Yellow Pippin Tag AL-G Orch-111	1	Row 38: Newtown Pippen (1)	R3-C13	Tag AL-G Orch-111 (Grandpa's Orchard, M-111) matches the workbook.
59	Apple — Pink Lady Tag AL-Pink/ AL-Pink Lady	2	Row 39: Pink Lady (Cripps Pink) (2)	R2-C7 , R3-C5	
60	Apple — Sweet Sixteen Tag AL- S Six	1	Row 41: Sweet Sixteen (2)	R5-C5	1 of 2 purchased found.
61	Apple — Ukn Rootstock Tag AL-Rtsk Unk	1	No purchase record matched	Graft first; reserved row R1 holds 13 of them	Rootstock of unknown type, for grafting.
62	Apple — Unknown Tag AL-UnKn	6	No purchase record matched	R3-C7 , R3-C12 , R4-C2 , R4-C5 , R4-C8 , R4-C10	Tag AL-UnKn. Purchased apples not otherwise found at Prairie: Braeburn 1, Cortland 2, Dolgo 1, Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Some of these 6 may be those trees; identify them at bloom or harvest.
63	Apple — Urban Gold Tag AL-UrbGld	1	Row 6: Blushing Delight Urban Columnar (1)	R5-C10	Urban Gold is matched to Blushing Delight, the yellow Urban apple that was purchased (row 6). The workbook's Golden Treat Urban (row 23) has QTY 0; confirm the label.
64	Apple — Urban Green Tag AL-UrbGrn	1	Row 42: Tandgy green Urban Columnar (1)	R4-C9	
65	Apple — Urban Red Tag AL-UrbR	1	Row 43: Tasty Red Urban Columnar (1)	R5-C9	
66	Apple — Zestar Tag AL-Zestar	2	Row 45: Zestar (3)	R4-C6 , R5-C8	2 of 3 purchased found.
67	Apricot — Chinese (Mormon) Tag AT-Chinese	1	Row 46: Chinese (Mormon) (1)	Not at Prairie (Boise)	Still in Boise (girdled, mulched); not planned for Prairie.
68	Apricot — Giant Tilton Tag AT-Tilton	1	Row 47: Giant Tilton (1)	R11-C3	
69	Apricot — Harcot Tag AT-Harcot	1	Row 48: Harcot (1)	R12-C1	
70	Apricot — Moore Park Tag AT-moorpark	2	Row 49: Moorpark (1)	R11-C1 , R12-C2	2 found, but 1 Moorpark was purchased. Patterson (row 50) was not found; one of these may be Patterson.

Inventory row	Prairie inventory	Qty	Matched purchase record(s)	Plan / positions	Notes
71	Apricot — Unknown Tag AT-Unk	1	No purchase record matched	R11-C2	Tag AT-Unk. The purchased Patterson (row 50) was not otherwise found.
72	Cherry — Bing Tag CR-Bing	2	Row 53: Bing (1); Row 54: Bing (1)	R9-C1 , R9-C3	
73	Cherry — Black Tartarian Tag CR-Black T	2	Row 55: Black Tartarian (3); Row 56: Black Tartarian (1)	R10-C1 , R10-C3	2 of 4 purchased found.
74	Cherry — Craigs Crimson Tag CR-Crimson	1	Row 57: Craig's Crimson (1)	Not at Prairie (Boise)	Still in Boise (girdled, mulched); not planned for Prairie.
75	Cherry — Hardy Giant Antique Tag CR-Hardy Antq	1	Row 58: Hardy Giant Antque Sweet (1)	R9-C7	
76	Cherry — Lapins Sweet Tag CR-Lapins	1	Row 59: Lapins (1)	R10-C2	
77	Cherry — Montmorency Tag CR-Mont	2	Row 60: Montmorency (2)	R10-C5 , R10-C8	
78	Cherry — Nanking, Ian Tag CR-Ian Nan	2	Row 61: Nanking (bush) (3)	R9-C5 , R9-C8	Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes.
79	Cherry — Nanking, Jules Tag CR-Juls Nan	1	Row 61: Nanking (bush) (3)	R10-C6	Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes.
80	Cherry — Royalton Sweet Tag CR-Royalton	1	Row 63: Royalton Sweet (1)	R9-C2	
81	Cherry — Sweet Tag CR-Sweet	3	No purchase record matched	R9-C6 , R10-C4 , R10-C7	Tag CR-Sweet. Purchased sweet cherries not otherwise found at Prairie: Black Tartarian 2, Regina 1, Sam 1.
82	Cherry — Sweet Stella Tag CR-Stella	1	Row 65: Stella (1)	R9-C4	
83	Peach — Blazing Star Tag PC-Blaz Star	1	Row 77: BlazingStar (1)	R12-C11	
84	Peach — Canadian Harmony Tag PC-Can Harm	1	Row 78: Canadian Harmony (1)	R11-C12	
85	Peach — Contender Tag PC-Contender	1	Row 79: Contender (2)	R11-C8	1 of 2 purchased found.
86	Peach — Elberta Tag PC-Elberta	1	Row 80: Elberta (2)	R12-C13	1 of 2 purchased found.
87	Peach — Frost Tag PC-Frost	1	Row 81: Frost (1)	R11-C11	
88	Peach — Hale Tag PC-Hale	1	Row 83: JH Hale (1)	R11-C10	
89	Peach — Redhaven Tag PC-Rd Haven	1	Row 84: Redhaven (1)	R11-C9	
90	Peach — Reliance Tag PC-Reliance	2	Row 85: Reliance (1); Row 86: Reliance (1); Row 87: Reliance (1)	R11-C13 , R12-C9	2 of 3 purchased found.
91	Peach — Unknown Tag PC-Unk	2	No purchase record matched	R12-C10 , R12-C12	Tag PC-Unk. Purchased peaches not otherwise found at Prairie: Contender 1, Elberta 1, Reliance 1, Veteran 1.
92	Pear — Anjou Stan. Supreme Tag PR-Anjou	1	Row 89: Anjou (2)	R7-C5	Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased.
93	Pear — Asian Hosui Tag PR-Asian H	1	Row 96: Hosui - Asian (1)	R8-C2	
94	Pear — Asian, Hardy Giant Tag PR-Asian	1	Row 105: Starking Hardy Giant - Asian (1)	R8-C1	
95	Pear — Asian, Shinseiki Tag PR-Shin Asian	1	Row 103: Shinseiki - Asian (1)	R7-C2	
96	Pear — Barnet Euro Tag PR-Barnet	1	Row 90: Barnett European Perry Pear (1)	R7-C7	

Inventory row	Prairie inventory	Qty	Matched purchase record(s)	Plan / positions	Notes
97	Pear — Bartlett Tag PR-Bart	2	Row 91: Bartlett (1); Row 92: Bartlett (1)	R8-C5, R8-C11	
98	Pear — Bartlett, Red Tag PR-R Bart	1	Row 102: Red Bartlett (1)	R7-C4	
99	Pear — Butt Euro Perry Tag PR-Butt Euro	1	Row 93: Butt European Perry Pear (1)	R7-C13	
100	Pear — Comice Tag PR-Comice	3	Row 94: Comice (3)	R7-C6, R7-C11, R8-C12	
101	Pear — D'Anjou Tag PR-D'an	2	Row 89: Anjou (2)	R8-C7, R8-C10	Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased.
102	Pear — Delicious Tag PR-Delic	1	Row 104: Starking Delicious (1)	R8-C9	
103	Pear — Hendre Huffeap Tag PR-Hendre	2	Row 95: Hendre Huffcapp European Perry Pear (1)	R8-C4, R8-C8	2 found, but 1 was purchased. Thorn (row 107) was not found; one of these may be Thorn.
104	Pear — Kieffer Tag PR-Kief	1	Row 97: Keiffer (1)	R7-C12	
105	Pear — Moonglow Tag PR-Moon	1	Row 98: Moonglow (1)	R8-C6	
106	Pear — Olympic Giant Tag PR-Olympic	1	Row 99: Olympic Giant - Asian (1)	R7-C3	
107	Pear — Parker Tag PR-Parker	1	Row 100: Parker (1)	R7-C9	
108	Pear — Pineapple Tag PR-Pineap	1	Row 101: Pineapple (1)	R8-C3	
109	Pear — Summercrisp Tag PR-Sum Cris	2	Row 106: SummerCrisp (2)	R7-C8, R7-C10	
110	Pear — Yellow Huffcapp Tag PR- Yw Huffcapp	1	Row 108: Yellow Huffcapp European Perry Pear (1)	R7-C1	
111	Plum — Italian Lovell Tag PM- It Lov	3	Row 109: Italian Plum (2); Row 110: Italian Seedling (2)	R11-C7, R12-C6, R12-C7	3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings.
112	Hican Tag hican	10	Row 69: Hican (10)	Not in the main orchard (nut tree)	
113	Pecan	4	Row 74: Pecan (Native Hardy) (10)	Not in the main orchard (nut tree)	4 of 10 purchased found.
114	Walnut — English Tag WL-Enligsh	4	No purchase record matched	Not in the main orchard (nut tree)	Not on the Fruit sheet. The inventory notes 3 are already planted.
115	Walnut — Heartnut Japanese Tag WL-Heart J	2	Row 68: Heartnut (2)	Not in the main orchard (nut tree)	

Also at Prairie but outside this plan. Berry bushes: Blackberry (various) 3, Hardy Kiwi 4, Sea Berry 11, Raspberry? 1. Perennials: Comfrey 25, Mint (various) 18, Rhubarb 15. Evergreens: Fir (Douglas) 2, Juniper (Eastern Red Ceder) 7, Juniper (Variety) 8, Pine (Austrian Black) 7, Pine (Ponderosa) 18, Pine (White) 2 (owner reports both died), Spruce (Blue) 4. Shade trees: Ash 3, Aspen (Trembling) 15, Maple (Sugar) 10, Maple 1, Oak (various) 8.

Apple Flowering-Group Research — September 24, 2026

Research covers the apple records that were missing flowering groups plus Harrison, mirrored from the dated notes in the Trees.xlsx F Group cells. Numbered assignments generally follow Orange Pippin; Foxwhelp follows Raintree, its recorded vendor; seed-grown Kerr uses an owner-approved best guess. Flowering groups are not universally standardized, and local bloom overlap still needs checking.

Apple / inventory record	Trees at Prairie	Workbook FG	Planning bloom / FG	Evidence, limitations, and sources
Blushing Delight Urban Columnar Fruit sheet row 6; R5-C10	1	Not recorded	Midseason; FG ?	Raintree describes the Urban Apple series as midseason bloomers and recommends different Urban cultivars for one another. No numbered flowering group was verified for Blushing Delight. Raintree: Blushing Delight
Campfield Cider Apple Fruit sheet row 9; R5-C1	1	3	FG 3	Orange Pippin explicitly assigns Campfield to flowering group 3. Raintree independently lists bloom group 3 (midseason). Both refer to flowering, not harvest season. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Orange Pippin: Campfield (FG 3) Raintree: Campfield (bloom group 3)

Apple / inventory record	Trees at Prairie	Workbook FG	Planning bloom / FG	Evidence, limitations, and sources
Dolgo Crabapple Fruit sheet row 13; R4-C13	1	2	FG 2	Orange Pippin explicitly assigns the named Dolgo cultivar to flowering group 2 and describes early blossom. This assignment assumes these are the named cultivar, not unnamed seedlings. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Orange Pippin: Dolgo
Foxwhelp Cider Apple Fruit sheet row 15; R2-C1	1	3	FG 3	The workbook records RainTree as the vendor for this tree, supporting the owner-selected Raintree group 3 (midseason). Raintree identifies its stock as Sonoma/Geneva Foxwhelp (Fauxwhelp), not authenticated English Foxwhelp. Other sources describe different Foxwhelp material (Orange Pippin group 5; RV Roger group B); those classifications are not substituted for the recorded supplier. Confirm local bloom. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Raintree: Fauxwhelp stock (bloom group 3) Orange Pippin: generic Foxwhelp entry (group 5; identity caveat) RV Roger: Foxwhelp (lettered group B)
Frequin Rouge Cider Apple Fruit sheet row 17; R2-C13	1	4	FG 4*	Orange Pippin assigns Frequin Rouge to group 4, corroborated by Raintree (mid-late). Cummins instead lists group 5 in its five-group system and warns that the precise Frequin Rouge variant in North America is uncertain. Use 4 as the source-attributed planning value, not proof of local bloom or a reason to substitute another Frequin cultivar. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Orange Pippin: Frequin Rouge (FG 4) Raintree: Frequin Rouge (bloom group 4) Cummins: Frequin Rouge (group 5; variant uncertainty) Cummins: five-group bloom scheme
Harrison Fruit sheet row 26; R4-C12, R5-C11	2	3	FG 3*	Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom. Orange Pippin: Harrison (FG 3) Raintree: Harrison (bloom group 4)
Kerr Crab Seedling Fruit sheet row 33; R3-C1, R5-C13, R6-C1	3	2	FG 2?	BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Orange Pippin: named Kerr cultivar (FG 2 only for the cultivar) Iowa State Extension: apple seedlings do not come true to type
Tandgy green Urban Columnar Fruit sheet row 42; R4-C9	1	Not recorded	Midseason; FG ?	The workbook spelling "Tandgy green" is treated as Tangy Green Urban Apple (AK-98); confirm the nursery label. Raintree gives midseason bloom guidance for the series and recommends other Urban cultivars. No numbered flowering group was verified. Raintree: Tangy Green
Tasty Red Urban Columnar Fruit sheet row 43; R5-C9	1	Not recorded	Midseason; FG ?	Raintree explicitly describes Urban Apple varieties as midseason bloomers and recommends them for one another. No numbered flowering group was verified for Tasty Red. Raintree: Tasty Red

Pollination Layout Check

Counts use the same screening as the partner lists below. Self-fertile trees are not counted as needing cross-pollination. Unidentified trees and trees without bloom or compatibility data have no screened partner yet; proximity alone is not counted as evidence.

Fruit type	Rows	Trees	Need cross-pollination	With a screened candidate ≤ 50 ft	With 2 distinct candidates ≤ 50 ft	Triploid/disputed with 2 mutual donors ≤ 50 ft	No screened partner yet	Adjacent same-cultivar pairs
Apples	R1–R6	54	48	42 of 48	42 of 48	5 of 5	6	0
Pears	R7–R8	25	21	16 of 21	12 of 21	—	5	0
Cherries	R9–R10	16	12	5 of 12	3 of 12	—	7	0
Peaches	R11–R12	11	3	1 of 3	1 of 3	—	2	0
Apricots	R11–R12	5	1	0 of 1	0 of 1	—	1	0
Plums	R11–R12	3	0	0 of 0	0 of 0	—	0	3

Pollination Design Notes

All 114 fruit trees found at Prairie are placed once, honoring each inventory quantity. Each fruit type has its own rows except R11–R12, which the apricots, plums, and peaches share. Open positions (68 in all) are at the east ends, between the blocks in the shared rows, and across R13–R15, which are held for trees you buy or graft later; they are not additional trees.

Row layouts: apples in 5 rows (13 + 13 + 13 + 13 + 2) plus reserved R1 (13 spots); pears in 2 rows (13 + 12); cherries in 2 rows (8 + 8); apricots at the west end (3 + 2), plums in the center (1 + 2), and peaches at the east end (6 + 5) share R11–R12; R13–R15 are open for trees you buy or graft later. Single-type blocks start at the west end, and a type's shorter row sits against its fuller rows to keep partners close.

Within each type, positions were chosen by a search that uses the same screening rules shown on this page. It favors two distinct candidates within 50 ft, two mutually screened diploid donors for triploid and disputed apples, and separation of copies of the same cultivar. It is a screened arrangement, not a guarantee of the best possible one.

Apple candidates use the workbook pollinizer lists and same/adjacent flowering groups, supplemented by the linked bloom research. The three Urban apples also use Raintree's within-series recommendation. Known triploids, disputed Liberty, unidentified trees, and unverified pollen donors are excluded as suppliers. Fuji/Aztec/Auvil, Honeycrisp/Super Red, and Bartlett/Red Bartlett count as shared cultivar identities.

Pear bloom charts are a screening tool, not proof of cross-compatibility. Asian pears are grouped without assuming European pears will pollinate them, and perry pears with different bloom windows are not treated as interchangeable.

Sweet cherries use compatibility groups plus same/adjacent bloom periods. Lapins is the only identified sweet cherry within one bloom period of both Black Tartarian and Bing, so the sweet cherries are grouped around it. Stella is not relied on for Bing. Montmorency is a self-fertile tart cherry, and the Nanking bushes are kept together and are not sweet-cherry donors.

J.H. Hale needs another peach. With Veteran missing, it sits beside identified non-Elberta peaches, following UC Davis guidance; unidentified peaches are not counted because one could be Elberta.

12 unidentified trees (6 apples, 3 sweet cherries, 2 peaches, 1 apricot) are placed among identified trees of their type but are not counted as donors until identified.

21 trees that are not self-fertile still have no screened partner, mostly unidentified trees and trees missing bloom or compatibility data. They are marked unconfirmed below; proximity alone is not counted as pollination evidence.

Row	Fruit type	Trees	Open	Placement notes
R1	Apple (reserved)	0 (13 reserved)	0	Reserved for 13 grafted rootstock apples; the inventory has 15 rootstocks (2 EMLA 7, 12 M-111, 1 unknown). Choose scion varieties whose bloom overlaps the apples in the next rows.
R2	Apple	13	0	Planning bloom: FG 3, FG 4, FG 4*.
R3	Apple	13	0	Planning bloom: FG 2?, FG 3, FG 4, FG ?.
R4	Apple	13	0	Planning bloom: FG 1, FG 2, FG 3, FG 3*, FG 4, FG ?, Midseason; FG ?.
R5	Apple	13	0	Planning bloom: FG 1, FG 2, FG 2?, FG 3, FG 3*, Midseason; FG ?.
R6	Apple	2	11	Planning bloom: FG 2?, FG 3.
R7	Pear	13	0	European/perry bloom windows and the Asian group checked separately.
R8	Pear	12	1	European/perry bloom windows and the Asian group checked separately.
R9	Cherry	8	5	Sweet cherries grouped around Lapins; tart and Nanking kept out of the sweet-cherry network.
R10	Cherry	8	5	Sweet cherries grouped around Lapins; tart and Nanking kept out of the sweet-cherry network.
R11	Apricot, plum, and peach (shared)	10	3	Apricots C1–C3 at the west end, plum C7 in the center, and peaches C8–C13 at the east end. Compact block of self-fertile apricots. Italian plums, self-fertile. J.H. Hale beside non-Elberta peaches; the others are self-fertile.
R12	Apricot, plum, and peach (shared)	9	4	Apricots C1–C2 at the west end, plums C6–C7 in the center, and peaches C9–C13 at the east end. Compact block of self-fertile apricots. Italian plums, self-fertile. J.H. Hale beside non-Elberta peaches; the others are self-fertile.
R13	Future trees	0	13	Held open for trees you buy or graft onto your rootstock later. Apples grafted here would be 140 ft from the nearest apples (R6), so plant at least two cultivars with overlapping bloom together.
R14	Future trees	0	13	Held open for trees you buy or graft onto your rootstock later. Apples grafted here would be 160 ft from the nearest apples (R6), so plant at least two cultivars with overlapping bloom together.
R15	Future trees	0	13	Held open for trees you buy or graft onto your rootstock later. Apples grafted here would be 180 ft from the nearest apples (R6), so plant at least two cultivars with overlapping bloom together.

Priority Pollination Pairings

Trees that are not self-fertile, plus triploid or disputed apples. Candidates are screened by fruit type, bloom timing, and recorded compatibility; distances are between tree centers.

Tree needing pollen	Position	Nearby candidates	Evidence / limitation
Foxwhelp Cider Apple	R2-C1	Braeburn (R2-C2, 18.0 ft) - Bloom-group candidate; Cortland (R3-C2, 21.9 ft) - Workbook-listed + bloom group	The workbook records RainTree as the vendor for this tree, supporting the owner-selected Raintree group 3 (midseason). Raintree identifies its stock as Sonoma/Geneva Foxwhelp (Fauxwhelp), not authenticated English Foxwhelp. Other sources describe different Foxwhelp material (Orange Pippin group 5; RV Roger group B); those classifications are not substituted for the recorded supplier. Confirm local bloom. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.
Jonagold	R2-C3	Braeburn (R2-C2, 18.0 ft) - Bloom-group candidate; Fuji (R2-C4, 18.0 ft) - Workbook-listed + bloom group	Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required.
Fuji	R2-C4	Honeycrisp (R2-C5, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R3-C5, 21.9 ft) - Workbook-listed + bloom group	Confirm overlapping bloom locally before relying on these candidates.
Honeycrisp	R2-C5	Cortland (R2-C6, 18.0 ft) - Bloom-group candidate; Fuji (R2-C4, 18.0 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Cortland	R2-C6	Honeycrisp (R2-C5, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R2-C7, 18.0 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Pink Lady (Cripps Pink)	R2-C7	Cortland (R2-C6, 18.0 ft) - Bloom-group candidate; Honeycrisp (R3-C8, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Jonagold	R2-C8	Gala (R2-C9, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R2-C7, 18.0 ft) - Workbook-listed + bloom group	Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required.
Arkansas Black Spur (FT015)	R2-C10	Cox's Orange Pippin (R2-C11, 18.0 ft) - Bloom-group candidate; Gala (R2-C9, 18.0 ft) - Bloom-group candidate	Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required.
Franklin Cider	R2-C12	Cox's Orange Pippin (R2-C11, 18.0 ft) - Workbook-listed + bloom group; Newtown Pippin (R3-C13, 21.9 ft) - Workbook-listed + bloom group	Confirm overlapping bloom locally before relying on these candidates.
Frequin Rouge Cider Apple	R2-C13	Franklin Cider (R2-C12, 18.0 ft) - Bloom-group candidate; Newtown Pippin (R3-C13, 21.9 ft) - Workbook-listed + bloom group	Orange Pippin assigns Frequin Rouge to group 4, corroborated by Raintree (mid-late). Cummins instead lists group 5 in its five-group system and warns that the precise Frequin Rouge variant in North America is uncertain. Use 4 as the source-attributed planning value, not proof of local bloom or a reason to substitute another Frequin cultivar. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.
Kerr Crab Seedling	R3-C1	Cortland (R3-C2, 18.0 ft) - Bloom-group candidate; provisional flowering group; Cox's Orange Pippin (R4-C1, 21.9 ft) - Bloom-group candidate; provisional flowering group	BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.
Cortland	R3-C2	Kingston Black (R3-C3, 18.0 ft) - Workbook-listed + bloom group; Braeburn (R2-C2, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Kingston Black	R3-C3	Cortland (R3-C2, 18.0 ft) - Bloom-group candidate; Braeburn (R2-C2, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Mutsu Crispin	R3-C4	Kingston Black (R3-C3, 18.0 ft) - Bloom-group candidate; Pink Lady (Cripps Pink) (R3-C5, 18.0 ft) - Bloom-group candidate	Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required.
Pink Lady (Cripps Pink)	R3-C5	Aztec Fuji (R3-C6, 18.0 ft) - Bloom-group candidate; Golden Delicious (R4-C4, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Aztec Fuji	R3-C6	Pink Lady (Cripps Pink) (R3-C5, 18.0 ft) - Bloom-group candidate; Cortland (R2-C6, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Unidentified apple	R3-C7	Unconfirmed — no screened partner in the available data.	Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.
Honeycrisp	R3-C8	Aztec Fuji (R3-C9, 18.0 ft) - Bloom-group candidate; Pink Lady (Cripps Pink) (R2-C7, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Aztec Fuji	R3-C9	GoldRush (R3-C10, 18.0 ft) - Bloom-group candidate; Honeycrisp (R3-C8, 18.0 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
GoldRush	R3-C10	Red Delicious (R3-C11, 18.0 ft) - Bloom-group candidate; Aztec Fuji (R3-C9, 18.0 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Red Delicious	R3-C11	GoldRush (R3-C10, 18.0 ft) - Bloom-group candidate; Cox's Orange Pippin (R2-C11, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.

Tree needing pollen	Position	Nearby candidates	Evidence / limitation
Unidentified apple	R3-C12	Unconfirmed — no screened partner in the available data.	Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.
Unidentified apple	R4-C2	Unconfirmed — no screened partner in the available data.	Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.
Honeycrisp (Super Red)	R4-C3	Golden Delicious (R4-C4, 18.0 ft) - Bloom-group candidate; Hudson's Golden Gem (R5-C3, 21.9 ft) - Workbook-listed + bloom group	Confirm overlapping bloom locally before relying on these candidates.
Golden Delicious	R4-C4	Honeycrisp (Super Red) (R4-C3, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R3-C5, 21.9 ft) - Workbook-listed + bloom group	Confirm overlapping bloom locally before relying on these candidates.
Unidentified apple	R4-C5	Unconfirmed — no screened partner in the available data.	Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.
Zestar	R4-C6	Wickson Crab (R4-C7, 18.0 ft) - Bloom-group candidate; Macintosh (R5-C7, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Wickson Crab	R4-C7	Zestar (R4-C6, 18.0 ft) - Bloom-group candidate; Macintosh (R5-C7, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Unidentified apple	R4-C8	Unconfirmed — no screened partner in the available data.	Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.
Tandgy green Urban Columnar	R4-C9	Blushing Delight Urban Columnar (R5-C10, 21.9 ft) - Raintree Urban Apple recommendation; confirm local bloom; Tasty Red Urban Columnar (R5-C9, 21.9 ft) - Raintree Urban Apple recommendation; confirm local bloom	The workbook spelling "Tandgy green" is treated as Tangy Green Urban Apple (AK-98); confirm the nursery label. Raintree gives midseason bloom guidance for the series and recommends other Urban cultivars. No numbered flowering group was verified.
Unidentified apple	R4-C10	Unconfirmed — no screened partner in the available data.	Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.
Auvil Early Fuji	R4-C11	Harrison (R4-C12, 18.0 ft) - Bloom-group candidate; CrimsonCrisp (R5-C12, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Harrison	R4-C12	Auvil Early Fuji (R4-C11, 18.0 ft) - Bloom-group candidate; CrimsonCrisp (R5-C12, 21.9 ft) - Bloom-group candidate	Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom.
Dolgo Crabapple	R4-C13	Harrison (R4-C12, 18.0 ft) - Workbook-listed + bloom group; Newtown Pippin (R3-C13, 21.9 ft) - Workbook-listed + bloom group	Orange Pippin explicitly assigns the named Dolgo cultivar to flowering group 2 and describes early blossom. This assignment assumes these are the named cultivar, not unnamed seedlings. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.
Campfield Cider Apple	R5-C1	Idared (R5-C2, 18.0 ft) - Bloom-group candidate; Cox's Orange Pippin (R4-C1, 21.9 ft) - Workbook-listed + bloom group	Orange Pippin explicitly assigns Campfield to flowering group 3. Raintree independently lists bloom group 3 (midseason). Both refer to flowering, not harvest season. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.
Idared	R5-C2	Hudson's Golden Gem (R5-C3, 18.0 ft) - Workbook-listed + bloom group; Cox's Orange Pippin (R4-C1, 21.9 ft) - Workbook-listed + bloom group	Confirm overlapping bloom locally before relying on these candidates.
Hudson's Golden Gem	R5-C3	Idared (R5-C2, 18.0 ft) - Bloom-group candidate; Granny Smith (R6-C2, 21.9 ft) - Workbook-listed + bloom group	Confirm overlapping bloom locally before relying on these candidates.
Liberty	R5-C4	Hudson's Golden Gem (R5-C3, 18.0 ft) - Bloom-group candidate; Sweet Sixteen (R5-C5, 18.0 ft) - Bloom-group candidate	Liberty ploidy/donor status is disputed. Excluded as a donor; given two diploid candidates.
Sweet Sixteen	R5-C5	Cortland (R5-C6, 18.0 ft) - Bloom-group candidate; Golden Delicious (R4-C4, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Cortland	R5-C6	Macintosh (R5-C7, 18.0 ft) - Bloom-group candidate; Sweet Sixteen (R5-C5, 18.0 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Macintosh	R5-C7	Cortland (R5-C6, 18.0 ft) - Bloom-group candidate; Zestar (R5-C8, 18.0 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Zestar	R5-C8	Macintosh (R5-C7, 18.0 ft) - Bloom-group candidate; Wickson Crab (R4-C7, 21.9 ft) - Bloom-group candidate	Confirm overlapping bloom locally before relying on these candidates.
Tasty Red Urban Columnar	R5-C9	Blushing Delight Urban Columnar (R5-C10, 18.0 ft) - Raintree Urban Apple recommendation; confirm local bloom; Tandgy green Urban Columnar (R4-C9, 21.9 ft) - Raintree Urban Apple recommendation; confirm local bloom	Raintree explicitly describes Urban Apple varieties as midseason bloomers and recommends them for one another. No numbered flowering group was verified for Tasty Red.
Blushing Delight Urban Columnar	R5-C10	Harrison (R5-C11, 18.0 ft) - Workbook-listed; confirm bloom; Tasty Red Urban Columnar (R5-C9, 18.0 ft) - Raintree Urban Apple recommendation; confirm local bloom	Raintree describes the Urban Apple series as midseason bloomers and recommends different Urban cultivars for one another. No numbered flowering group was verified for Blushing Delight.
Harrison	R5-C11	CrimsonCrisp (R5-C12, 18.0 ft) - Bloom-group candidate; Auvil Early Fuji (R4-C11, 21.9 ft) - Bloom-group candidate	Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom.

Tree needing pollen	Position	Nearby candidates	Evidence / limitation
CrimsonCrisp	R5-C12	Harrison (R5-C11, 18.0 ft) - Workbook-listed + bloom group; Auvil Early Fuji (R4-C11, 21.9 ft) - Workbook-listed + bloom group	Confirm overlapping bloom locally before relying on these candidates.
Kerr Crab Seedling	R5-C13	CrimsonCrisp (R5-C12, 18.0 ft) - Bloom-group candidate; provisional flowering group; Harrison (R4-C12, 21.9 ft) - Bloom-group candidate; provisional flowering group	BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.
Kerr Crab Seedling	R6-C1	Granny Smith (R6-C2, 18.0 ft) - Bloom-group candidate; provisional flowering group; Idared (R5-C2, 21.9 ft) - Bloom-group candidate; provisional flowering group	BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.
Yellow Huffcapp European Perry Pear	R7-C1	Unconfirmed — no screened partner in the available data.	Yellow Huffcap bloom and compatibility need confirmation; do not assume all perry pears overlap.
Shinseiki - Asian	R7-C2	Unconfirmed — no screened partner in the available data.	Shinseiki is grouped near the other Asian pears; a named compatible pair remains unconfirmed.
Olympic Giant - Asian	R7-C3	Unconfirmed — no screened partner in the available data.	Confirm Asian-pear nursery identity and bloom. The two Giant selections are not counted as independent partners.
Red Bartlett	R7-C4	Anjou (R7-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Hendre Huffcapp European Perry Pear (R8-C4, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Anjou	R7-C5	Comice (R7-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Red Bartlett (R7-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Barnett European Perry Pear	R7-C7	Unconfirmed — no screened partner in the available data.	Confirm whether Barnett is Barnet. Barnet is very late in the perry chart; no partner is assumed.
SummerCrisp	R7-C8	Parker (R7-C9, 18.0 ft) - Nursery-recommended Parker for Summercrisp; confirm local bloom	Confirm overlapping bloom locally before relying on these candidates.
Parker	R7-C9	Unconfirmed — no screened partner in the available data.	Parker is recommended for Summercrisp; the reverse pairing was not verified.
SummerCrisp	R7-C10	Parker (R7-C9, 18.0 ft) - Nursery-recommended Parker for Summercrisp; confirm local bloom	Confirm overlapping bloom locally before relying on these candidates.
Keiffer	R7-C12	Comice (R7-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Butt European Perry Pear (R7-C13, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Butt European Perry Pear	R7-C13	Keiffer (R7-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Comice (R8-C12, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility	Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility.
Starking Hardy Giant - Asian	R8-C1	Hosui - Asian (R8-C2, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm Asian-pear nursery identity and bloom. The two Giant selections are not counted as independent partners.
Hosui - Asian	R8-C2	Starking Hardy Giant - Asian (R8-C1, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Hendre Huffcapp European Perry Pear	R8-C4	Pineapple (R8-C3, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility.
Bartlett	R8-C5	Hendre Huffcapp European Perry Pear (R8-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Moonglow	R8-C6	Anjou (R8-C7, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Anjou	R8-C7	Hendre Huffcapp European Perry Pear (R8-C8, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Hendre Huffcapp European Perry Pear	R8-C8	Anjou (R8-C7, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Starking Delicious (R8-C9, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility.
Starking Delicious	R8-C9	Anjou (R8-C10, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Hendre Huffcapp European Perry Pear (R8-C8, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Anjou	R8-C10	Starking Delicious (R8-C9, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Bartlett	R8-C11	Comice (R8-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Anjou (R8-C10, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	Confirm overlapping bloom locally before relying on these candidates.
Bing	R9-C1	Royalton Sweet (R9-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Lapins (R10-C2, 33.6 ft) - Cherry compatibility + bloom chart; confirm local overlap	Treat Bing as cross-pollination dependent. Stella is conservatively excluded as its donor.
Royalton Sweet	R9-C2	Bing (R9-C1, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Lapins (R10-C2, 21.9 ft) - Cherry compatibility + bloom chart; confirm local overlap	Royalton and Sam are incompatible, but Sam was not found at Prairie. Royalton uses Bing, Lapins, and Stella.
Bing	R9-C3	Royalton Sweet (R9-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Lapins (R10-C2, 21.9 ft) - Cherry compatibility + bloom chart; confirm local overlap	Treat Bing as cross-pollination dependent. Stella is conservatively excluded as its donor.

Tree needing pollen	Position	Nearby candidates	Evidence / limitation
Nanking (bush)	R9-C5	Unconfirmed — no screened partner in the available data.	Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines.
Unidentified sweet cherry	R9-C6	Unconfirmed — no screened partner in the available data.	Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified.
Hardy Giant Antque Sweet	R9-C7	Unconfirmed — no screened partner in the available data.	Sweet-cherry donor compatibility was not verified; do not rely on this tree to supply other varieties.
Nanking (bush)	R9-C8	Unconfirmed — no screened partner in the available data.	Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines.
Black Tartarian	R10-C1	Lapins (R10-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap	Black Tartarian blooms very early; Lapins is its only identified donor in an adjacent bloom period. Two of the four purchased Black Tartarians were not found and may be among the unidentified sweet cherries.
Black Tartarian	R10-C3	Lapins (R10-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap	Black Tartarian blooms very early; Lapins is its only identified donor in an adjacent bloom period. Two of the four purchased Black Tartarians were not found and may be among the unidentified sweet cherries.
Unidentified sweet cherry	R10-C4	Unconfirmed — no screened partner in the available data.	Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified.
Nanking (bush)	R10-C6	Unconfirmed — no screened partner in the available data.	Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines.
Unidentified sweet cherry	R10-C7	Unconfirmed — no screened partner in the available data.	Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified.
Unidentified apricot	R11-C2	Unconfirmed — no screened partner in the available data.	Unidentified apricot, possibly the purchased Patterson (self-fertile in the workbook). Confirm its identity.
JH Hale	R11-C10	Frost (R11-C11, 18.0 ft) - UC Davis: almost any non-Elberta peach; match chill and bloom; Redhaven (R11-C9, 18.0 ft) - UC Davis: almost any non-Elberta peach; match chill and bloom	J.H. Hale needs pollen from another peach, despite the workbook label. Veteran, its earlier partner, was not found at Prairie. UC Davis says almost any other peach works except Elberta and its close relatives, so Hale sits beside identified non-Elberta peaches; match chill and bloom.
Unidentified peach	R12-C10	Unconfirmed — no screened partner in the available data.	Unidentified peach, possibly Contender, Elberta, Reliance, or Veteran. Not counted as a J.H. Hale pollinizer until identified, because Elberta does not work for Hale.
Unidentified peach	R12-C12	Unconfirmed — no screened partner in the available data.	Unidentified peach, possibly Contender, Elberta, Reliance, or Veteran. Not counted as a J.H. Hale pollinizer until identified, because Elberta does not work for Hale.

Full Tree Inventory by Row

One record per planted tree, south (R1) to north and west (C1) to east. The tag is the inventory tag ID to look for in the field. Original workbook fertility values are retained; planning corrections and uncertainty are shown separately.

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R2-C1	FOX	Foxwhelp Cider Apple Tag AL-Foxwhelp; inventory name: Foxwhelp Cider	Apple	Semi Dwarf	FG 3	Not recorded	The workbook records RainTree as the vendor for this tree, supporting the owner-selected Raintree group 3 (midseason). Raintree identifies its stock as Sonoma/Geneva Foxwhelp (Fauxwhelp), not authenticated English Foxwhelp. Other sources describe different Foxwhelp material (Orange Pippin group 5; RV Roger group B); those classifications are not substituted for the recorded supplier. Confirm local bloom. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Braeburn (R2-C2, 18.0 ft) - Bloom-group candidate; Cortland (R3-C2, 21.9 ft) - Workbook-listed + bloom group Bloom research and sources	39 (15)
R2-C2	BB	Braeburn Tag AL-Brae Dwf; inventory name: Braeburn Dwarf	Apple	Not recorded (tag says Dwarf)	FG 4	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Cortland (R3-C2, 21.9 ft) - Workbook-listed + bloom group; Kingston Black (R3-C3, 21.9 ft) - Bloom-group candidate Inventory match: 1 of 2 purchased Braeburns found. The tag says Dwarf, so it is matched to the Lowes tree with no recorded rootstock (row 8) rather than the M-111 tree (row 7).	27 (8, 7)
R2-C3	JG	Jonagold Tag AL-Jon G; inventory name: Jonagold	Apple	Semi Dwarf	FG 4	Not self-fertile	Triploid Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required. Braeburn (R2-C2, 18.0 ft) - Bloom-group candidate; Fuji (R2-C4, 18.0 ft) - Workbook-listed + bloom group	53 (32)
R2-C4	FJ	Fuji Tag AL-Fuji; inventory name: Fuji	Apple	Semi Dwarf	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Honeycrisp (R2-C5, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R3-C5, 21.9 ft) - Workbook-listed + bloom group Inventory match: Same tag text (AL-Fuji) as the Early Fuji; check the variety label when planting.	42 (18)
R2-C5	HC	Honeycrisp Tag AL-Hon Cris; inventory name: Honeycrisp	Apple	M-111	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Cortland (R2-C6, 18.0 ft) - Bloom-group candidate; Fuji (R2-C4, 18.0 ft) - Bloom-group candidate Inventory match: 2 of 3 purchased found.	49 (28)
R2-C6	CL	Cortland Tag AL-Cort/ AL-Cortland; inventory name: Cortland	Apple	M-111	FG 3	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Honeycrisp (R2-C5, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R2-C7, 18.0 ft) - Bloom-group candidate Inventory match: 3 of 5 purchased found.	29 (10)
R2-C7	PL	Pink Lady (Cripps Pink) Tag AL-Pink/ AL-Pink Lady; inventory name: Pink Lady	Apple	Not recorded	FG 3	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Cortland (R2-C6, 18.0 ft) - Bloom-group candidate; Honeycrisp (R3-C8, 21.9 ft) - Bloom-group candidate	59 (39)
R2-C8	JG	Jonagold Tag AL-Jon G; inventory name: Jonagold	Apple	Semi Dwarf	FG 4	Not self-fertile	Triploid Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required. Gala (R2-C9, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R2-C7, 18.0 ft) - Workbook-listed + bloom group	53 (32)
R2-C9	GL	Gala Tag AL-Gala; inventory name: Gala	Apple	M-111	FG 4	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Aztec Fuji (R3-C9, 21.9 ft) - Workbook-listed + bloom group; GoldRush (R3-C10, 21.9 ft) - Workbook-listed + bloom group Inventory match: 1 of 2 purchased found.	45 (19, 20)
R2-C10	AB	Arkansas Black Spur (FT015) Tag AL-Ark Blk; inventory name: Arkansas Black	Apple	M-111	FG 3	Not self-fertile	Triploid Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required. Cox's Orange Pippin (R2-C11, 18.0 ft) - Bloom-group candidate; Gala (R2-C9, 18.0 ft) - Bloom-group candidate	26 (2)
R2-C11	CO	Cox's Orange Pippin Tag AL-O Pip; inventory name: Cox Orange Pippin	Apple	M-7	FG 3	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Franklin Cider (R2-C12, 18.0 ft) - Bloom-group candidate; Red Delicious (R3-C11, 21.9 ft) - Bloom-group candidate	30 (11)
R2-C12	FC	Franklin Cider Tag AL-Frank C; inventory name: Franklin Cider	Apple	Semi Dwarf	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Cox's Orange Pippin (R2-C11, 18.0 ft) - Workbook-listed + bloom group; Newtown Pippin (R3-C13, 21.9 ft) - Workbook-listed + bloom group	40 (16)
R2-C13	FRQ	Frequin Rouge Cider Apple Tag AL-Frequin; inventory name: Frequin Rouge	Apple	Semi Dwarf	FG 4*	Not recorded	Orange Pippin assigns Frequin Rouge to group 4, corroborated by Raintree (mid-late). Cummins instead lists group 5 in its five-group system and warns that the precise Frequin Rouge variant in North America is uncertain. Use 4 as the source-attributed planning value, not proof of local bloom or a reason to substitute another Frequin cultivar. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Franklin Cider (R2-C12, 18.0 ft) - Bloom-group candidate; Newtown Pippin (R3-C13, 21.9 ft) - Workbook-listed + bloom group Bloom research and sources	41 (17)

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R3-C1	KER	Kerr Crab Seedling Tag AL-Kerr Crab; inventory name: Crabapple, Kerr	Apple	Not recorded	FG 2?	Not recorded	BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Cortland (R3-C2, 18.0 ft) - Bloom-group candidate; provisional flowering group; Cox's Orange Pippin (R4-C1, 21.9 ft) - Bloom-group candidate; provisional flowering group Inventory match: 3 found, but 2 were purchased. All are seed-grown, so each may bloom differently. Bloom research and sources	32 (33)
R3-C2	CL	Cortland Tag AL-Cort/ AL-Cortland; inventory name: Cortland	Apple	M-111	FG 3	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Kingston Black (R3-C3, 18.0 ft) - Workbook-listed + bloom group; Braeburn (R2-C2, 21.9 ft) - Bloom-group candidate Inventory match: 3 of 5 purchased found.	29 (10)
R3-C3	KB	Kingston Black Tag AL-G Orch-7; inventory name: Kingston Black	Apple	M-7	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Cortland (R3-C2, 18.0 ft) - Bloom-group candidate; Braeburn (R2-C2, 21.9 ft) - Bloom-group candidate Inventory match: Tag AL-G Orch-7 (Grandpa's Orchard, M-7) matches the workbook.	54 (34)
R3-C4	MC	Mutsu Crispin Tag AL-Crisp; inventory name: Crispin	Apple	M-111	FG 3	Not self-fertile	Triploid Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required. Kingston Black (R3-C3, 18.0 ft) - Bloom-group candidate; Pink Lady (Cripps Pink) (R3-C5, 18.0 ft) - Bloom-group candidate	35 (37)
R3-C5	PL	Pink Lady (Cripps Pink) Tag AL-Pink/ AL-Pink Lady; inventory name: Pink Lady	Apple	Not recorded	FG 3	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Aztec Fuji (R3-C6, 18.0 ft) - Bloom-group candidate; Golden Delicious (R4-C4, 21.9 ft) - Bloom-group candidate	59 (39)
R3-C6	FA	Aztec Fuji Tag AL-Az Fuji; inventory name: Fuji Aztec	Apple	Semi Dwarf or Dwarf	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Pink Lady (Cripps Pink) (R3-C5, 18.0 ft) - Bloom-group candidate; Cortland (R2-C6, 21.9 ft) - Bloom-group candidate Inventory match: 2 of 3 purchased found (row 4: 1 semi-dwarf; row 5: 2 dwarf); the inventory does not say which.	43 (4, 5)
R3-C7	UNK	Unidentified apple Tag AL-UnKn; inventory name: Unknown	Apple	Not recorded	FG ?	Not recorded	Unidentified Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. Unconfirmed — no screened partner in the available data. Inventory match: Tag AL-UnKn. Purchased apples not otherwise found at Prairie: Braeburn 1, Cortland 2, Dolgo 1, Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Some of these 6 may be those trees; identify them at bloom or harvest.	62
R3-C8	HC	Honeycrisp Tag AL-Hon Cris; inventory name: Honeycrisp	Apple	M-111	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Aztec Fuji (R3-C9, 18.0 ft) - Bloom-group candidate; Pink Lady (Cripps Pink) (R2-C7, 21.9 ft) - Bloom-group candidate Inventory match: 2 of 3 purchased found.	49 (28)
R3-C9	FA	Aztec Fuji Tag AL-Az Fuji; inventory name: Fuji Aztec	Apple	Semi Dwarf or Dwarf	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. GoldRush (R3-C10, 18.0 ft) - Bloom-group candidate; Honeycrisp (R3-C8, 18.0 ft) - Bloom-group candidate Inventory match: 2 of 3 purchased found (row 4: 1 semi-dwarf; row 5: 2 dwarf); the inventory does not say which.	43 (4, 5)
R3-C10	GO	GoldRush Tag AL-Gld Rsh; inventory name: Goldrush	Apple	B-118	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Red Delicious (R3-C11, 18.0 ft) - Bloom-group candidate; Aztec Fuji (R3-C9, 18.0 ft) - Bloom-group candidate	46 (24)
R3-C11	RD	Red Delicious Tag AL-Red Del; inventory name: Delicious, Red	Apple	M-111	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. GoldRush (R3-C10, 18.0 ft) - Bloom-group candidate; Cox's Orange Pippin (R2-C11, 21.9 ft) - Bloom-group candidate	37 (40)
R3-C12	UNK	Unidentified apple Tag AL-UnKn; inventory name: Unknown	Apple	Not recorded	FG ?	Not recorded	Unidentified Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. Unconfirmed — no screened partner in the available data. Inventory match: Tag AL-UnKn. Purchased apples not otherwise found at Prairie: Braeburn 1, Cortland 2, Dolgo 1, Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Some of these 6 may be those trees; identify them at bloom or harvest.	62
R3-C13	NP	Newtown Pippin Tag AL-G Orch-111; inventory name: Newton Yellow Pippin	Apple	M-111	FG 3	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Franklin Cider (R2-C12, 21.9 ft) - Bloom-group candidate; Harrison (R4-C12, 21.9 ft) - Bloom-group candidate Inventory match: Tag AL-G Orch-111 (Grandpa's Orchard, M-111) matches the workbook.	58 (38)
R4-C1	CO	Cox's Orange Pippin Tag AL-O Pip; inventory name: Cox Orange Pippin	Apple	M-7	FG 3	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Cortland (R3-C2, 21.9 ft) - Bloom-group candidate; Idared (R5-C2, 21.9 ft) - Bloom-group candidate	30 (11)

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R4-C2	UNK	Unidentified apple Tag AL-UnKn; inventory name: Unknown	Apple	Not recorded	FG ?	Not recorded	Unidentified Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. Unconfirmed — no screened partner in the available data. Inventory match: Tag AL-UnKn. Purchased apples not otherwise found at Prairie: Braeburn 1, Cortland 2, Dolgo 1, Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Some of these 6 may be those trees; identify them at bloom or harvest.	62
R4-C3	HC	Honeycrisp (Super Red) Tag AL-Hny Crisp; inventory name: Honeycrisp, Super Red	Apple	M-7	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Golden Delicious (R4-C4, 18.0 ft) - Bloom-group candidate; Hudson's Golden Gem (R5-C3, 21.9 ft) - Workbook-listed + bloom group	50 (29)
R4-C4	GD	Golden Delicious Tag AL-Gold D; inventory name: Delicious, Golden	Apple	M-111	FG 4	Partly Self-fertile	★ Good donor Confirm overlapping bloom locally before relying on these candidates. Honeycrisp (Super Red) (R4-C3, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R3-C5, 21.9 ft) - Workbook-listed + bloom group	36 (21)
R4-C5	UNK	Unidentified apple Tag AL-UnKn; inventory name: Unknown	Apple	Not recorded	FG ?	Not recorded	Unidentified Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. Unconfirmed — no screened partner in the available data. Inventory match: Tag AL-UnKn. Purchased apples not otherwise found at Prairie: Braeburn 1, Cortland 2, Dolgo 1, Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Some of these 6 may be those trees; identify them at bloom or harvest.	62
R4-C6	ZT	Zestar Tag AL-Zestar; inventory name: Zestar	Apple	M-111	FG 1	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Wickson Crab (R4-C7, 18.0 ft) - Bloom-group candidate; Macintosh (R5-C7, 21.9 ft) - Bloom-group candidate Inventory match: 2 of 3 purchased found.	66 (45)
R4-C7	WC	Wickson Crab Tag AL-Wickson cr; inventory name: Crabapple, Wickson	Apple	B-9	FG 2	Not self-fertile	★ Good donor Confirm overlapping bloom locally before relying on these candidates. Zestar (R4-C6, 18.0 ft) - Bloom-group candidate; Macintosh (R5-C7, 21.9 ft) - Bloom-group candidate	33 (44)
R4-C8	UNK	Unidentified apple Tag AL-UnKn; inventory name: Unknown	Apple	Not recorded	FG ?	Not recorded	Unidentified Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. Unconfirmed — no screened partner in the available data. Inventory match: Tag AL-UnKn. Purchased apples not otherwise found at Prairie: Braeburn 1, Cortland 2, Dolgo 1, Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Some of these 6 may be those trees; identify them at bloom or harvest.	62
R4-C9	UG	Tandgy green Urban Columnar Tag AL-UrbGrn; inventory name: Urban Green	Apple	10x3	Midseason; FG ?	Not self-fertile	The workbook spelling "Tandgy green" is treated as Tangy Green Urban Apple (AK-98); confirm the nursery label. Raintree gives midseason bloom guidance for the series and recommends other Urban cultivars. No numbered flowering group was verified. Blushing Delight Urban Columnar (R5-C10, 21.9 ft) - Raintree Urban Apple recommendation; confirm local bloom; Tasty Red Urban Columnar (R5-C9, 21.9 ft) - Raintree Urban Apple recommendation; confirm local bloom Bloom research and sources	64 (42)
R4-C10	UNK	Unidentified apple Tag AL-UnKn; inventory name: Unknown	Apple	Not recorded	FG ?	Not recorded	Unidentified Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. Unconfirmed — no screened partner in the available data. Inventory match: Tag AL-UnKn. Purchased apples not otherwise found at Prairie: Braeburn 1, Cortland 2, Dolgo 1, Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Some of these 6 may be those trees; identify them at bloom or harvest.	62
R4-C11	AF	Auvil Early Fuji Tag AL-Fuji; inventory name: Fuji, Early	Apple	M-111	FG 4	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Harrison (R4-C12, 18.0 ft) - Bloom-group candidate; CrimsonCrisp (R5-C12, 21.9 ft) - Bloom-group candidate Inventory match: Same tag text (AL-Fuji) as the Fuji; check the variety label when planting.	44 (3)
R4-C12	HR	Harrison Tag AL-Harrison; inventory name: Harrison	Apple	M-111 or Semi Dwarf	FG 3*	Not self-fertile	Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom. Auvil Early Fuji (R4-C11, 18.0 ft) - Bloom-group candidate; CrimsonCrisp (R5-C12, 21.9 ft) - Bloom-group candidate Inventory match: The workbook lists one Harrison (row 26, M-111) and one Harrison Cider (row 27, semi-dwarf), the same cultivar. The tags do not say which is which, so both use the row 26 cultivar data. Bloom research and sources	48 (26, 27)

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R4-C13	DLG	Dolgo Crabapple Tag AL-Dolgo; inventory name: Crabapple, Dolgo	Apple	Not recorded	FG 2	Not recorded	Orange Pippin explicitly assigns the named Dolgo cultivar to flowering group 2 and describes early blossom. This assignment assumes these are the named cultivar, not unnamed seedlings. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Harrison (R4-C12, 18.0 ft) - Workbook-listed + bloom group; Newtown Pippin (R3-C13, 21.9 ft) - Workbook-listed + bloom group Inventory match: 1 of 2 purchased found. Bloom research and sources	31 (13)
R5-C1	CAM	Campfield Cider Apple Tag AL-Campfield; inventory name: Campfield Cide	Apple	Semi Dwarf	FG 3	Not recorded	Orange Pippin explicitly assigns Campfield to flowering group 3. Raintree independently lists bloom group 3 (midseason). Both refer to flowering, not harvest season. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Idared (R5-C2, 18.0 ft) - Bloom-group candidate; Cox's Orange Pippin (R4-C1, 21.9 ft) - Workbook-listed + bloom group Bloom research and sources	28 (9)
R5-C2	ID	Idared Tag AL-Idared; inventory name: Idared	Apple	M-111	FG 2	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Hudson's Golden Gem (R5-C3, 18.0 ft) - Workbook-listed + bloom group; Cox's Orange Pippin (R4-C1, 21.9 ft) - Workbook-listed + bloom group	52 (31)
R5-C3	HG	Hudson's Golden Gem Tag AL-Gold Gem; inventory name: Hudsons Golden Gem	Apple	M-111	FG 3	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Idared (R5-C2, 18.0 ft) - Bloom-group candidate; Granny Smith (R6-C2, 21.9 ft) - Workbook-listed + bloom group	51 (30)
R5-C4	LB	Liberty Tag AL-Liberty; inventory name: Liberty	Apple	M-111	FG 2	Not self-fertile	Disputed Liberty ploidy/donor status is disputed. Excluded as a donor; given two diploid candidates. Hudson's Golden Gem (R5-C3, 18.0 ft) - Bloom-group candidate; Sweet Sixteen (R5-C5, 18.0 ft) - Bloom-group candidate	55 (35)
R5-C5	SS	Sweet Sixteen Tag AL- S Six; inventory name: Sweet Sixteen	Apple	M-7	FG 3	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Cortland (R5-C6, 18.0 ft) - Bloom-group candidate; Golden Delicious (R4-C4, 21.9 ft) - Bloom-group candidate Inventory match: 1 of 2 purchased found.	60 (41)
R5-C6	CL	Cortland Tag AL-Cort/ AL-Cortland; inventory name: Cortland	Apple	M-111	FG 3	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Macintosh (R5-C7, 18.0 ft) - Bloom-group candidate; Sweet Sixteen (R5-C5, 18.0 ft) - Bloom-group candidate Inventory match: 3 of 5 purchased found.	29 (10)
R5-C7	MT	Macintosh Tag AL-Mc In; inventory name: McIntosh	Apple	Semi Dwarf	FG 2	semi-self fertile	Confirm overlapping bloom locally before relying on these candidates. Cortland (R5-C6, 18.0 ft) - Bloom-group candidate; Zestar (R5-C8, 18.0 ft) - Bloom-group candidate	57 (36)
R5-C8	ZT	Zestar Tag AL-Zestar; inventory name: Zestar	Apple	M-111	FG 1	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Macintosh (R5-C7, 18.0 ft) - Bloom-group candidate; Wickson Crab (R4-C7, 21.9 ft) - Bloom-group candidate Inventory match: 2 of 3 purchased found.	66 (45)
R5-C9	UR	Tasty Red Urban Columnar Tag AL-UrbR; inventory name: Urban Red	Apple	10x3 (MM106)	Midseason; FG ?	Not self-fertile	Raintree explicitly describes Urban Apple varieties as midseason bloomers and recommends them for one another. No numbered flowering group was verified for Tasty Red. Blushing Delight Urban Columnar (R5-C10, 18.0 ft) - Raintree Urban Apple recommendation; confirm local bloom; Tandgy green Urban Columnar (R4-C9, 21.9 ft) - Raintree Urban Apple recommendation; confirm local bloom Bloom research and sources	65 (43)
R5-C10	UD	Blushing Delight Urban Columnar Tag AL-UrbGld; inventory name: Urban Gold	Apple	10x3	Midseason; FG ?	Not self-fertile	Raintree describes the Urban Apple series as midseason bloomers and recommends different Urban cultivars for one another. No numbered flowering group was verified for Blushing Delight. Harrison (R5-C11, 18.0 ft) - Workbook-listed; confirm bloom; Tasty Red Urban Columnar (R5-C9, 18.0 ft) - Raintree Urban Apple recommendation; confirm local bloom Inventory match: Urban Gold is matched to Blushing Delight, the yellow Urban apple that was purchased (row 6). The workbook's Golden Treat Urban (row 23) has QTY 0; confirm the label. Bloom research and sources	63 (6)
R5-C11	HR	Harrison Tag AL-Harrison; inventory name: Harrison	Apple	M-111 or Semi Dwarf	FG 3*	Not self-fertile	Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom. CrimsonCrisp (R5-C12, 18.0 ft) - Bloom-group candidate; Auvil Early Fuji (R4-C11, 21.9 ft) - Bloom-group candidate Inventory match: The workbook lists one Harrison (row 26, M-111) and one Harrison Cider (row 27, semi-dwarf), the same cultivar. The tags do not say which is which, so both use the row 26 cultivar data. Bloom research and sources	48 (26, 27)
R5-C12	CC	CrimsonCrisp Tag AL-Crimson; inventory name: Crimson Crisp	Apple	M-111	FG 3	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Harrison (R5-C11, 18.0 ft) - Workbook-listed + bloom group; Auvil Early Fuji (R4-C11, 21.9 ft) - Workbook-listed + bloom group	34 (12)

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R5-C13	KER	Kerr Crab Seedling Tag AL-Kerr Crab; inventory name: Crabapple, Kerr	Apple	Not recorded	FG 2?	Not recorded	BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. CrimsonCrisp (R5-C12, 18.0 ft) - Bloom-group candidate; provisional flowering group; Harrison (R4-C12, 21.9 ft) - Bloom-group candidate; provisional flowering group Inventory match: 3 found, but 2 were purchased. All are seed-grown, so each may bloom differently. Bloom research and sources	32 (33)
R6-C1	KER	Kerr Crab Seedling Tag AL-Kerr Crab; inventory name: Crabapple, Kerr	Apple	Not recorded	FG 2?	Not recorded	BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. Granny Smith (R6-C2, 18.0 ft) - Bloom-group candidate; provisional flowering group; Idared (R5-C2, 21.9 ft) - Bloom-group candidate; provisional flowering group Inventory match: 3 found, but 2 were purchased. All are seed-grown, so each may bloom differently. Bloom research and sources	32 (33)
R6-C2	GS	Granny Smith Tag AL-Gran S; inventory name: Granny Smith	Apple	Semi Dwarf	FG 3	Self-fertile	Self-fertile★ Good donor Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Hudson's Golden Gem (R5-C3, 21.9 ft) - Bloom-group candidate; Idared (R5-C2, 21.9 ft) - Bloom-group candidate	47 (25)
R7-C1	YHC	Yellow Huffcapp European Perry Pear Tag PR- Yw Huffcapp; inventory name: Yellow Huffcapp	Pear	Semi Dwarf	Unconfirmed	Not recorded	Yellow Huffcap bloom and compatibility need confirmation; do not assume all perry pears overlap. Unconfirmed — no screened partner in the available data.	110 (108)
R7-C2	SA	Shinseiki - Asian Tag PR-Shin Asian; inventory name: Asian, Shinseiki	Pear	Semi Dwarf	Unconfirmed	Partly self-fertile	Shinseiki is grouped near the other Asian pears; a named compatible pair remains unconfirmed. Unconfirmed — no screened partner in the available data.	95 (103)
R7-C3	OG	Olympic Giant - Asian Tag PR-Olympic; inventory name: Olympic Giant	Pear	Standard	Unconfirmed	Partly self-fertile	Confirm Asian-pear nursery identity and bloom. The two Giant selections are not counted as independent partners. Unconfirmed — no screened partner in the available data.	106 (99)
R7-C4	RB	Red Bartlett Tag PR-R Bart; inventory name: Bartlett, Red	Pear	Not recorded	Early–mid	Not recorded	Confirm overlapping bloom locally before relying on these candidates. Anjou (R7-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Hendre Huffcapp European Perry Pear (R8-C4, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility	98 (102)
R7-C5	AU	Anjou Tag PR-Anjou; inventory name: Anjou Stan. Supreme	Pear	Standard	Early–mid	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Comice (R7-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Red Bartlett (R7-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility Inventory match: Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased.	92 (89)
R7-C6	CE	Comice Tag PR-Comice; inventory name: Comice	Pear	Semi Dwarf	Mid–late	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Anjou (R7-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility	100 (94)
R7-C7	BNT	Barnett European Perry Pear Tag PR-Barnet; inventory name: Barnet Euro	Pear	Semi Dwarf	Unconfirmed	Not recorded	Confirm whether Barnett is Barnet. Barnet is very late in the perry chart; no partner is assumed. Unconfirmed — no screened partner in the available data.	96 (90)
R7-C8	SMC	SummerCrisp Tag PR-Sum Cris; inventory name: Summercrisp	Pear	Not recorded	Unconfirmed	Not recorded	Confirm overlapping bloom locally before relying on these candidates. Parker (R7-C9, 18.0 ft) - Nursery-recommended Parker for Summercrisp; confirm local bloom	109 (106)
R7-C9	PKR	Parker Tag PR-Parker; inventory name: Parker	Pear	Not recorded	Unconfirmed	Not recorded	Parker is recommended for Summercrisp; the reverse pairing was not verified. Unconfirmed — no screened partner in the available data.	107 (100)
R7-C10	SMC	SummerCrisp Tag PR-Sum Cris; inventory name: Summercrisp	Pear	Not recorded	Unconfirmed	Not recorded	Confirm overlapping bloom locally before relying on these candidates. Parker (R7-C9, 18.0 ft) - Nursery-recommended Parker for Summercrisp; confirm local bloom	109 (106)
R7-C11	CE	Comice Tag PR-Comice; inventory name: Comice	Pear	Semi Dwarf	Mid–late	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Keiffer (R7-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Anjou (R8-C10, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility	100 (94)

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R7-C12	KF	Keiffer Tag PR-Kief; inventory name: Kieffer	Pear	Not recorded	Late	Not recorded	Confirm overlapping bloom locally before relying on these candidates. Comice (R7-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Butt European Perry Pear (R7-C13, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	104 (97)
R7-C13	BUT	Butt European Perry Pear Tag PR-Butt Euro; inventory name: Butt Euro Perry	Pear	Semi Dwarf	Late (perry chart)	Not recorded	Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility. Keiffer (R7-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Comice (R8-C12, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility	99 (93)
R8-C1	HG	Starking Hardy Giant - Asian Tag PR-Asian; inventory name: Asian, Hardy Giant	Pear	Semi Dwarf	Early	Not self-fertile	Confirm Asian-pear nursery identity and bloom. The two Giant selections are not counted as independent partners. Hosui - Asian (R8-C2, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	94 (105)
R8-C2	HA	Hosui - Asian Tag PR-Asian H; inventory name: Asian Hosui	Pear	Standard	Early	Not self-fertile	Confirm overlapping bloom locally before relying on these candidates. Starking Hardy Giant - Asian (R8-C1, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	93 (96)
R8-C3	PE	Pineapple Tag PR-Pineap; inventory name: Pineapple	Pear	Semi Dwarf	Early	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Hendre Huffcapp European Perry Pear (R8-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Red Bartlett (R7-C4, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility	108 (101)
R8-C4	HHC	Hendre Huffcapp European Perry Pear Tag PR-Hendre; inventory name: Hendre Huffeap	Pear	Semi Dwarf	Early (perry chart)	Not recorded	Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility. Pineapple (R8-C3, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility Inventory match: 2 found, but 1 was purchased. Thorn (row 107) was not found; one of these may be Thorn.	103 (95)
R8-C5	BT	Bartlett Tag PR-Bart; inventory name: Bartlett	Pear	Standard	Early–mid	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Hendre Huffcapp European Perry Pear (R8-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	97 (91, 92)
R8-C6	MW	Moonglow Tag PR-Moon; inventory name: Moonglow	Pear	Standard	Early–mid	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Anjou (R8-C7, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	105 (98)
R8-C7	AU	Anjou Tag PR-D'an; inventory name: D'Anjou	Pear	Standard	Early–mid	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Hendre Huffcapp European Perry Pear (R8-C8, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility Inventory match: Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased.	101 (89)
R8-C8	HHC	Hendre Huffcapp European Perry Pear Tag PR-Hendre; inventory name: Hendre Huffeap	Pear	Semi Dwarf	Early (perry chart)	Not recorded	Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility. Anjou (R8-C7, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Starking Delicious (R8-C9, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility Inventory match: 2 found, but 1 was purchased. Thorn (row 107) was not found; one of these may be Thorn.	103 (95)
R8-C9	SD	Starking Delicious Tag PR-Delic; inventory name: Delicious	Pear	Standard	Early–mid	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Anjou (R8-C10, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Hendre Huffcapp European Perry Pear (R8-C8, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	102 (104)
R8-C10	AU	Anjou Tag PR-D'an; inventory name: D'Anjou	Pear	Standard	Early–mid	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Starking Delicious (R8-C9, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility Inventory match: Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased.	101 (89)
R8-C11	BT	Bartlett Tag PR-Bart; inventory name: Bartlett	Pear	Standard	Early–mid	Partly self-fertile	Confirm overlapping bloom locally before relying on these candidates. Comice (R8-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Anjou (R8-C10, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility	97 (91, 92)
R8-C12	CE	Comice Tag PR-Comice; inventory name: Comice	Pear	Semi Dwarf	Mid–late	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Bartlett (R8-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Keiffer (R7-C12, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility	100 (94)
R9-C1	BB	Bing Tag CR-Bing; inventory name: Bing	Cherry	Semi Dwarf	Cherry period 3	Partly self-fertile	Pseudomonas + Treat Bing as cross-pollination dependent. Stella is conservatively excluded as its donor. Royalton Sweet (R9-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Lapins (R10-C2, 33.6 ft) - Cherry compatibility + bloom chart; confirm local overlap Pseudomonas positive, untreated; 1 of the 2 trees with this tag.	72 (53, 54)

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R9-C2	RS	Royalton Sweet Tag CR-Royalton; inventory name: Royalton Sweet	Cherry	Semi Dwarf	Cherry period 3	Partly self-fertile	Pseudomonas + Royalton and Sam are incompatible, but Sam was not found at Prairie. Royalton uses Bing, Lapins, and Stella. Bing (R9-C1, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Lapins (R10-C2, 21.9 ft) - Cherry compatibility + bloom chart; confirm local overlap Pseudomonas positive, untreated.	80 (63)
R9-C3	BB	Bing Tag CR-Bing; inventory name: Bing	Cherry	Semi Dwarf	Cherry period 3	Partly self-fertile	Pseudomonas + Treat Bing as cross-pollination dependent. Stella is conservatively excluded as its donor. Royalton Sweet (R9-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Lapins (R10-C2, 21.9 ft) - Cherry compatibility + bloom chart; confirm local overlap Pseudomonas positive, untreated; 1 of the 2 trees with this tag.	72 (53, 54)
R9-C4	ST	Stella Tag CR-Stella; inventory name: Sweet Stella	Cherry	Semi Dwarf	Cherry period 3	self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. Bing (R9-C3, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Lapins (R10-C2, 33.6 ft) - Cherry compatibility + bloom chart; confirm local overlap Pseudomonas positive, untreated.	82 (65)
R9-C5	NKI	Nanking (bush) Tag CR-Ian Nan; inventory name: Nanking, Ian	Cherry	Not recorded	Unconfirmed	Not recorded	Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines. Unconfirmed — no screened partner in the available data. Inventory match: Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes.	78 (61)
R9-C6	SWT	Unidentified sweet cherry Tag CR-Sweet; inventory name: Sweet	Cherry	Not recorded	Unconfirmed	Not recorded	Unidentified Pseudomonas + Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified. Unconfirmed — no screened partner in the available data. Inventory match: Tag CR-Sweet. Purchased sweet cherries not otherwise found at Prairie: Black Tartarian 2, Regina 1, Sam 1. Pseudomonas positive, untreated; 1 of the 3 trees with this tag.	81
R9-C7	HG	Hardy Giant Antque Sweet Tag CR-Hardy Antq; inventory name: Hardy Giant Antique	Cherry	Semi Dwarf	Unconfirmed	Partly self-fertile	Pseudomonas + Sweet-cherry donor compatibility was not verified; do not rely on this tree to supply other varieties. Unconfirmed — no screened partner in the available data. Pseudomonas positive, untreated.	75 (58)
R9-C8	NKI	Nanking (bush) Tag CR-Ian Nan; inventory name: Nanking, Ian	Cherry	Not recorded	Unconfirmed	Not recorded	Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines. Unconfirmed — no screened partner in the available data. Inventory match: Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes.	78 (61)
R10-C1	BT	Black Tartarian Tag CR-Black T; inventory name: Black Tartarian	Cherry	Standard	Cherry period 1	Partly self-fertile	Pseudomonas + Black Tartarian blooms very early; Lapins is its only identified donor in an adjacent bloom period. Two of the four purchased Black Tartarians were not found and may be among the unidentified sweet cherries. Lapins (R10-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap Inventory match: 2 of 4 purchased found. Pseudomonas positive, untreated; 1 of the 2 trees with this tag.	73 (55, 56)
R10-C2	LP	Lapins Tag CR-Lapins; inventory name: Lapins Sweet	Cherry	Not recorded	Cherry period 2	Partly self-fertile	Self-fertile Lapins is self-fertile in the reference chart and is the only identified sweet cherry within one bloom period of both Black Tartarian and Bing, so the sweet cherries are grouped around it. Black Tartarian (R10-C1, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Bing (R9-C3, 21.9 ft) - Cherry compatibility + bloom chart; confirm local overlap	76 (59)
R10-C3	BT	Black Tartarian Tag CR-Black T; inventory name: Black Tartarian	Cherry	Standard	Cherry period 1	Partly self-fertile	Pseudomonas + Black Tartarian blooms very early; Lapins is its only identified donor in an adjacent bloom period. Two of the four purchased Black Tartarians were not found and may be among the unidentified sweet cherries. Lapins (R10-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap Inventory match: 2 of 4 purchased found. Pseudomonas positive, untreated; 1 of the 2 trees with this tag.	73 (55, 56)
R10-C4	SWT	Unidentified sweet cherry Tag CR-Sweet; inventory name: Sweet	Cherry	Not recorded	Unconfirmed	Not recorded	Unidentified Pseudomonas + Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified. Unconfirmed — no screened partner in the available data. Inventory match: Tag CR-Sweet. Purchased sweet cherries not otherwise found at Prairie: Black Tartarian 2, Regina 1, Sam 1. Pseudomonas positive, untreated; 1 of the 3 trees with this tag.	81

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R10-C5	MNT	Montmorency Tag CR-Mont; inventory name: Montmorency	Cherry	Not recorded	Unconfirmed	Not recorded	Self-fertile Montmorency is a self-fertile tart cherry; not used in the sweet-cherry donor network. No extra partner required under the current self-fertility guidance.	77 (60)
R10-C6	NKJ	Nanking (bush) Tag CR-Juls Nan; inventory name: Nanking, Jules	Cherry	Not recorded	Unconfirmed	Not recorded	Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines. Unconfirmed — no screened partner in the available data. Inventory match: Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes.	79 (61)
R10-C7	SWT	Unidentified sweet cherry Tag CR-Sweet; inventory name: Sweet	Cherry	Not recorded	Unconfirmed	Not recorded	Unidentified Pseudomonas + Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified. Unconfirmed — no screened partner in the available data. Inventory match: Tag CR-Sweet. Purchased sweet cherries not otherwise found at Prairie: Black Tartarian 2, Regina 1, Sam 1. Pseudomonas positive, untreated; 1 of the 3 trees with this tag.	81
R10-C8	MNT	Montmorency Tag CR-Mont; inventory name: Montmorency	Cherry	Not recorded	Unconfirmed	Not recorded	Self-fertile Montmorency is a self-fertile tart cherry; not used in the sweet-cherry donor network. No extra partner required under the current self-fertility guidance.	77 (60)
R11-C1	MP	Moorpark Tag AT-moorpark; inventory name: Moore Park	Apricot	Not recorded	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Inventory match: 2 found, but 1 Moorpark was purchased. Patterson (row 50) was not found; one of these may be Patterson. Pseudomonas positive, untreated; 1 of the 2 trees with this tag.	70 (49)
R11-C2	UNK	Unidentified apricot Tag AT-Unk; inventory name: Unknown	Apricot	Not recorded	Unconfirmed	Not recorded	Unidentified Pseudomonas + Unidentified apricot, possibly the purchased Patterson (self-fertile in the workbook). Confirm its identity. Unconfirmed — no screened partner in the available data. Inventory match: Tag AT-Unk. The purchased Patterson (row 50) was not otherwise found. Pseudomonas positive, untreated.	71
R11-C3	GT	Giant Tilton Tag AT-Tilton; inventory name: Giant Tilton	Apricot	Standard	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Pseudomonas positive, untreated.	68 (47)
R11-C7	IPL	Italian Plum Tag PM- It Lov; inventory name: Italian Lovell	Plum	Not recorded	Unconfirmed	Not recorded	Self-fertile Tag "Italian Lovell": Italian plum, probably on Lovell rootstock. The named Italian plum is self-fertile or partly self-fertile in references; if one of these is a purchased Italian seedling, its fertility is unknown. No extra partner required under the current self-fertility guidance. Inventory match: 3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings.	111 (109, 110)
R11-C8	CR	Contender Tag PC-Contender; inventory name: Contender	Peach	Standard	Unconfirmed	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Inventory match: 1 of 2 purchased found.	85 (79)
R11-C9	RH	Redhaven Tag PC-Rd Haven; inventory name: Redhaven	Peach	Semi Dwarf	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Pseudomonas positive, untreated.	89 (84)
R11-C10	JH	JH Hale Tag PC-Hale; inventory name: Hale	Peach	Semi Dwarf	Unconfirmed	Self-fertile	Pseudomonas + J.H. Hale needs pollen from another peach, despite the workbook label. Veteran, its earlier partner, was not found at Prairie. UC Davis says almost any other peach works except Elberta and its close relatives, so Hale sits beside identified non-Elberta peaches; match chill and bloom. Frost (R11-C11, 18.0 ft) - UC Davis: almost any non-Elberta peach; match chill and bloom; Redhaven (R11-C9, 18.0 ft) - UC Davis: almost any non-Elberta peach; match chill and bloom Pseudomonas positive, untreated.	88 (83)
R11-C11	FT	Frost Tag PC-Frost; inventory name: Frost	Peach	Semi Dwarf	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Pseudomonas positive, untreated.	87 (81)

Position	Code	Variety and tag	Fruit type	Rootstock	Bloom / FG	Fertility (workbook)	Planning notes / nearest candidates	Inventory row (Fruit row)
R11-C12	CH	Canadian Harmony Tag PC-Can Harm; inventory name: Canadian Harmony	Peach	Semi Dwarf	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Pseudomonas positive, untreated.	84 (78)
R11-C13	RC	Reliance Tag PC-Reliance; inventory name: Reliance	Peach	Standard	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Inventory match: 2 of 3 purchased found. Pseudomonas positive, untreated.	90 (85, 86, 87)
R12-C1	HT	Harcot Tag AT-Harcot; inventory name: Harcot	Apricot	Semi Dwarf	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Pseudomonas positive, untreated.	69 (48)
R12-C2	MP	Moorpark Tag AT-moorpark; inventory name: Moore Park	Apricot	Not recorded	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Inventory match: 2 found, but 1 Moorpark was purchased. Patterson (row 50) was not found; one of these may be Patterson. Pseudomonas positive, untreated; 1 of the 2 trees with this tag.	70 (49)
R12-C6	IPL	Italian Plum Tag PM- It Lov; inventory name: Italian Lovell	Plum	Not recorded	Unconfirmed	Not recorded	Self-fertile Tag "Italian Lovell": Italian plum, probably on Lovell rootstock. The named Italian plum is self-fertile or partly self-fertile in references; if one of these is a purchased Italian seedling, its fertility is unknown. No extra partner required under the current self-fertility guidance. Inventory match: 3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings.	111 (109, 110)
R12-C7	IPL	Italian Plum Tag PM- It Lov; inventory name: Italian Lovell	Plum	Not recorded	Unconfirmed	Not recorded	Self-fertile Tag "Italian Lovell": Italian plum, probably on Lovell rootstock. The named Italian plum is self-fertile or partly self-fertile in references; if one of these is a purchased Italian seedling, its fertility is unknown. No extra partner required under the current self-fertility guidance. Inventory match: 3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings.	111 (109, 110)
R12-C9	RC	Reliance Tag PC-Reliance; inventory name: Reliance	Peach	Standard	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Inventory match: 2 of 3 purchased found. Pseudomonas positive, untreated.	90 (85, 86, 87)
R12-C10	UNK	Unidentified peach Tag PC-Unk; inventory name: Unknown	Peach	Not recorded	Unconfirmed	Not recorded	Unidentified Pseudomonas + Unidentified peach, possibly Contender, Elberta, Reliance, or Veteran. Not counted as a J.H. Hale pollinizer until identified, because Elberta does not work for Hale. Unconfirmed — no screened partner in the available data. Inventory match: Tag PC-Unk. Purchased peaches not otherwise found at Prairie: Contender 1, Elberta 1, Reliance 1, Veteran 1. Pseudomonas positive, untreated.	91
R12-C11	BS	BlazingStar Tag PC-Blaz Star; inventory name: Blazing Star	Peach	Not recorded	Unconfirmed	Self-fertile	Self-fertile Pseudomonas + Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Pseudomonas positive, untreated.	83 (77)
R12-C12	UNK	Unidentified peach Tag PC-Unk; inventory name: Unknown	Peach	Not recorded	Unconfirmed	Not recorded	Unidentified Pseudomonas + Unidentified peach, possibly Contender, Elberta, Reliance, or Veteran. Not counted as a J.H. Hale pollinizer until identified, because Elberta does not work for Hale. Unconfirmed — no screened partner in the available data. Inventory match: Tag PC-Unk. Purchased peaches not otherwise found at Prairie: Contender 1, Elberta 1, Reliance 1, Veteran 1. Pseudomonas positive, untreated.	91
R12-C13	EA	Elberta Tag PC-Elberta; inventory name: Elberta	Peach	Standard	Unconfirmed	Self-fertile	Self-fertile Self-fertility is useful insurance; local bloom and insect activity still affect cropping. No extra partner required under the current self-fertility guidance. Inventory match: 1 of 2 purchased found.	86 (80)

Not in the Main Orchard

Nut trees are kept out of the main orchard. Rootstocks need grafting before they can be planted as fruit trees, and the Boise trees were not moved to Prairie. Windbreak, east-hedge, and berry guild plants are covered in the windbreak section and the site plan notes.

Inventory row	Item	Qty	Status and notes
38	Apple — EMLA7 Apple Rootstock · Tag AL-Rtsk-EMLA7	2	Rootstock: graft first, then plant in reserved row R1. Ungrafted EMLA 7 rootstock, for grafting.
56	Apple — M 111 Rootstock · Tag AL-Rstk M111	12	Rootstock: graft first, then plant in reserved row R1. Ungrafted M-111 rootstock, for grafting.
61	Apple — Ukn Rootstock · Tag AL-Rtsk Unk	1	Rootstock: graft first, then plant in reserved row R1. Rootstock of unknown type, for grafting.
67	Apricot — Chinese (Mormon) · Tag AT-Chinese	1	Still in Boise (girdled, mulched); not planned for Prairie.
74	Cherry — Craigs Crimson · Tag CR-Crimson	1	Still in Boise (girdled, mulched); not planned for Prairie.
112	Hican · Tag hican	10	Nut tree: kept out of the main orchard.
113	Pecan	4	Nut tree: kept out of the main orchard. 4 of 10 purchased found.
114	Walnut — English · Tag WL-Enligsh	4	Nut tree: kept out of the main orchard. Not on the Fruit sheet. The inventory notes 3 are already planted.
115	Walnut — Heartnut Japanese · Tag WL-Heart J	2	Nut tree: kept out of the main orchard.

Penn State notes that black walnut and butternut are the best-known sources of juglone, a compound that injures many plants, and that English walnut and hickories (the group that includes pecans and hicans) also produce it to a lesser degree. Heartnut is a walnut closely related to butternut. Site these nut trees away from the orchard rather than beside it.

Once grafted, 13 of the 15 rootstocks go in reserved row R1; the other 2 can fill open spots at the east end of R6, the short apple row (11 open). R13–R15 are held open for more trees you buy or graft later.

The Chinese apricot and Craig's Crimson cherry are still in Boise (girdled and mulched) and are not part of the Prairie plan.

Planning Corrections and Sources

Inventory_PrairiePlants.xlsx decides which trees are planned; the Trees.xlsx Fruit sheet supplies cultivar data. Every inventory line is matched to its purchase record(s) in the reconciliation table. When one inventory line matches several records of the same cultivar (for example Harrison or Aztec Fuji), the first record's cultivar data is used and the rootstock lists the possibilities. The plot width, east-side spacing, windbreak, east hedge, juniper distance, and white pine losses are owner updates recorded in this page, not in the workbooks. Berry bushes are matched to the Trees.xlsx Berry sheet.

Invalid cross-species spreadsheet entries are not used: Bartlett/Moonglow/Anjou are not cherry pollenizers, and Jefferson/Theta are not pear pollenizers. The size note in the Asian Hardy Giant pollenizer cell is not a pollination relationship.

Planning corrections are separate from the source fields: Liberty donor status remains disputed; Lapins and Montmorency are treated as self-fertile; Bing is not treated as reliably self-fertile; J.H. Hale needs another peach despite its workbook label, and Elberta is not used for it. Original fertility values remain visible in the inventory.

Apple flowering-group research is dated September 24, 2026 and linked in its own table. Workbook FG values drive the grid colors, bloom labels, and candidate screening. Seed-grown Kerr uses group 2 only as an owner-approved best guess (FG 2?). Qualitative Urban Apple bloom seasons remain unnumbered.

Pseudomonas (bacterial canker) results come from the inventory. Positive, untreated trees are flagged but not relocated; the layout is driven by pollination.

Before planting, confirm cultivar identities, bloom overlap, local climate suitability, rootstock and mature size, and insect activity. Windbreak sizes and spacings are planning values from the cited guides; adjust them to the plants you have.

1. Flowering groups and overlapping bloom — Orange Pippin

2. Fruit-tree pollination, cultivar identity, and cherry/plum cautions — Cummins Nursery

3. Apple cross-compatibility and pollenizers — Washington State University

4. Liberty: disputed ploidy and pollenizing ability — Orange Pippin

5. Pear bloom and pollination chart — Stark Bro’s

6. Perry-pear flowering observations — National Perry Pear Centre

7. Summercrisp: Parker named as a pollenizer — Charlie the Tree Guy

8. Sweet-cherry compatibility and bloom chart — Oregon State University

9. J.H. Hale: almost any other peach except Elberta; Veteran named — UC Davis Foundation Plant Services

10. J.H. Hale pollinizers and the Elberta exception — Ask Extension
11. Windbreak row spacing, in-row spacing, end extension, and protection distance — University of Nebraska–Lincoln EC1767

12. Hybrid hazelnut growth habit, spacing, hand harvest, and coppicing — Upper Midwest Hazelnut Development Initiative

13. Hazelnut seedlings ripening late August into October — UMHDI harvest timing bulletin

14. Hybrid hazelnuts in windbreaks: taller tree rows at least 15 ft away — USDA National Agroforestry Center

15. Theta as a Jefferson hazelnut pollenizer — Oregon State University

16. Elderberry spacing and hardwood cuttings — West Virginia University Extension

17. Currant and gooseberry spacing, sun, self-fertility, and jostaberry rust resistance — University of Minnesota Extension

18. White pine blister rust: five-needle pines and currant/gooseberry hosts — WSU HortSense
19. Honeyberries: partial shade and two varieties for pollination — University of Alaska Fairbanks Cooperative Extension

20. Comfrey: deep roots make it difficult to eradicate — WSU HortSense

21. Red mulberry size, male/female trees, and pruning time — Missouri Botanical Garden

22. Red mulberry: a tree, root sprouting, and crosses with white mulberry — USDA Forest Service (FEIS)

23. Red vs. white mulberry: open-grown form and western windbreaks — Purdue Extension FNR-237

24. Harvesting mulberries by shaking branches over a tarp — Ask Extension

25. Mulberry pruning: cuts on limbs 2 in or thicker heal poorly — Ask Extension

26. Cedar-apple rust and keeping junipers away from apples — University of Minnesota Extension

27. Juglone and nut trees — Penn State Extension

Prairie Plants Orchard — Inventory-based planting plan, October 4, 2026 — 237 × 600 ft orchard plot plus 35 ft west windbreak, C1 berry guilds, and east elderberry hedge — 13 across — 18 × 20 ft triangular offset — Planning schematic; verify site suitability, identities, and bloom before planting.